DEPARTMENT OF MICROBIOLOGY

M.Sc., Microbiology

REGULATIONS AND SYLLABUS

[For the candidates admitted from the Academic Year 2022 – 2023 onwards]

ALAGAPPA UNIVERSITY
(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC)
Karaikudi -630003, Tamil Nadu.

The panel of Members-Broad Based Board of Studies

Chairperson: **Name**: Dr. A. Arun, **Designation**: Professor and Head, Department: Microbiology, **University**: Alagappa University, **Teaching Experience:** 20years, **Research Experience:** 20years, **Area of Research:** Bioenergy, Bioremediation and Bioplastics	
Foreign Expert (Industry) : Name Dr. Sudhakar Muniyasamy, Designation: Senior Scientist and Technical leader for Bioplastics and Biodegradable polymers, Department: Chemicals Cluster - Advanced Polymer Composite Ressearch, University- CSIR, South Africa, Research Experience:15 yeras, Area ofResearch: Bioplastics.	
Indian Expert: **Name**: Dr. R. Thirumurugan, **Designation:** Professor **Department**: Animal Science, **University:** Bharathidasan University, Trichy.	
Indian Expert: Name: Dr. V. Rajesh Kannan, **Designation:** Professor, **Department**: Microbiology, **University:** Bharathidasan University, Trichy.	
INDUSTRY EXPERT: NAME: D. Suresh Lingam, **Designation**: Managing Director, Yaazh Genomics, Madurai	
Member: Name: Dr. T. Kavitha, **Designation**: Assistant Professor **Department:** Microbiology, **University: Alagappa University** **Teaching Experience**: 15years, **ResearchExperience**: 7years, **Area of Research**: Agricultural Microbiology, Environmental Microbiology	

Member: Name: Dr. T. Sathiyamoorthi, **Designation**: Assistant Professor **Department:** Microbiology, **University:** Alagappa University **Teaching Experience**: 7years, **ResearchExperience**: 7years, **Area of Research**: Medical Microbiology	
Member: Name: Dr. V.Balasubramaniyan, **Designation:** Assistant Professor, **Department:** Microbiology, **University:** Alagappa University **Teaching Experience**: 6years, **Research Experience**: 13years, Area of Research:	
Alumnus Name: P. Priyanka, **Current position:** Lab technician, **Professional address:** MSK computerized lab, Karaikudi	

ALAGAPPA UNIVERSITY
DEPARTMENT OF MICROBIOLOGY
Karaikudi -630003, Tamil Nadu.

REGULATIONS AND SYLLABUS- (CBCS-University Department)
[For the candidates admitted from the Academic Year 2022 – 2023 onwards]

Name of the Department: **Department of Microbiology**

Name of the Subject Discipline: **Microbiology**

Programme of Level: **M.Sc.,**
Duration for the Course: Full Time (Two Years)

1. Choice-Based Credit System

A choice-Based Credit System is a flexible system of learning. This system allows students togain knowledge at their own tempo. Students shall decide on electives from a wide range of elective courses offered by the University Departments in consultation with the Department committee. As a result, students undergo additional courses and acquire more than the required number of credits. They can also adopt an inter-disciplinary and intra-disciplinary approach to learning and make the best use of available faculty expertise.

2. Programme

"Programme" means a course of study leading to the award of a degree in a discipline.

3. Courses

'Course' is a component (a paper) of a programme. A unique course code identifies each course offered by the Department. A course contains lectures/ tutorials/laboratory/seminar/project / practical training/report writing /Viva-voce, etc., or a combination of these to effectively meet the teaching and learning needs.

4. Credits

The term "Credit" refers to the weightage given to a course, usually in relation to the instructional hours assigned to it. Normally in each of the courses, credits will be assigned on the basis of the number of lectures/tutorials/laboratory and other forms of learning required to complete the course contents in a 15-week schedule. One credit is equal to one hour of lecture per week. For laboratory/field work, one credit is equal to two hours.

5. Semesters

An Academic year is divided into two Semesters**.** Each semester, courses are offered in 15 teaching weeks, and the remaining 5 weeks are to be utilized for examination and evaluation purposes. Each week has 30 working hours spread over 5 days a week.

6. Medium of Instruction

All the courses will be instructed in English

7. Departmental committee

The Departmental Committee consists of the faculty of the Department. The Departmental Committee shall be responsible for admission to all the programs offered by the Department, including the conduct of entrance tests, verification of records, admission, and evaluation. The Departmental Committee determines the course deliberation and specifies the allocation of credits semester and course-wise. Each course will also identify the number of credits for lectures, tutorials, practical, seminars, etc. The courses(Core/Discipline Specific Elective/Non-Major Elective) are designed by teachers and approved by the Departmental Committees. The Board shall approve courses approved by the Departmental Committees of Studies/Broad-Based Board of Studies. A teacher offering a course will also be responsible for maintaining attendance and performance sheets (CIA -I, CIA-II, assignments, and seminar) of all the students registered for the course. The Non-major elective programme, MOOCs coordinator, and Internship Mentor are responsible for submitting the performance sheet to the Head of the department. The Head of the Department consolidates all such performance sheets of courses pertaining to the programmes offered by the department. Then forward the same to be Controller of Examinations.

8. Programme General Objectives- (PGO)

PGO-1	To know the basics of microbes
PGO-2	To learn the various industrial application of microorganisms
PGO-3	To understand the concepts of production of recombinant vaccines and other pharmaceutical products using microbes
PGO-4	To inculcate the students about environmental cleanup by microbes
PGO-5	To make the learner aspirant in research
PGO-6	To understand the importance of indispensable living creature

9. Programme Specific Objectives-(PSO)

PSO-1	To acquire knowledge of principles of the microbial world
PSO-2	To make the students understand on fundamental interaction of the microbes with other biological and inanimate elements
PSO-3	To understand the rationale in the field of Applied Microbiology
PSO-4	To enable the students technically sound in the Microbial Techniques
PSO-5	To produce the student with more research knowledge on Recent Trends in Microbiology
PSO-6	To make them experts in spheres of Microbiology

10. Programme Outcome-(PO)

PO-1	Understand the concepts of microbiology with relevant structure and diversity
PO-2	Acquire knowledge on biological molecules and metabolism necessary for future research
PO-3	Get ideas on the current application of microorganism human welfare through rDNA technology
PO-4	Become expertise in the field of microbiology, both in theoretical and practical aspects.
PO-5	Obtain knowledge on the benefits of sustainable metabolite production from microbes that replace synthetic products.
PO-6	Students become a skilled person in all spheres of microbiology

11. Eligibility for admission

A candidate who has passed Bachelor's Degree in Biological Sciences (Microbiology, Biochemistry, Biotechnology, Botany, Zoology, Bioinformatics, Agricultural / Veterinary / Fisheries Sciences / Pharmacy) degree with at least 50% of marks and 45% marks for SC/ST candidates as main course of study of any university accepted by the syndicate as equivalent thereto, subject to such condition as may be prescribed therefore shall be permitted to appear and qualify for the M.Sc. Degree in Microbiology of this University after a course of study of two academic years.

12. Minimum Duration of programme

The programme is for a period of two years. Each year shall consist of two semesters, viz. Odd and Even semesters. Odd semesters shall be from June / July to October / November, and even semesters shall be from November / December to April / May. Each semester there shall be 90 working days consisting of 6 teaching hours per working day (5 days/week).

13. Components

A PG programme consists of a number of courses. The term "course" is applied to indicate a logicalpart of the program's subject matter. It is invariably equivalent to a "paper" subject matter in the conventional sense. The following are the various categories of the courses suggestedfor the PG programmes:

A. Core courses (CC)- "Core Papers" means "the core courses" related to the programme concerned, including practicals and project work offered under the programme, and shall cover core competency, critical thinking, analytical reasoning, and research skill.

B. Discipline-Specific Electives (DSE) means the courses offered under the programme related to the major but are to be selected by the students shall cover additional academic knowledge, critical thinking, and analytical reasoning.

C. Non-Major Electives (NME)- Exposure beyond the discipline

- Students have to undergo a total of two Non- Major Elective courses with 2 credits offered by other departments (one in II Semester and another in III Semester).
- A uniform time frame of 3 hours on a common day (Tuesday) shall be allocated for the Non-Major Electives.
- Non-Major Elective courses offered by the departments pertaining to a semester should be announced before the end of the previous semester.
- Registration process: Students must register for the Non-Major Elective course within 15 days from the commencement of the semester, either in the department or NME portal (University Website).

D. Self Learning Courses from MOOCs platforms.

- MOOCs shall be voluntary for the students.
- Students must undergo 2 Self LearningCourses (MOOCs), one in the 2nd semester and another in the 3rd semester.
  - The actual credits earned through MOOCs shall be transferred to the credit plan of programs as extra credits. Otherwise 2 credits/course be given if the Self Learning Course (MOOCs) is without credit.
- While selecting the MOOCs, preference shall be given to the course related to employability skills.

E. Internship

The students have to go for an Internship. The student can undergo industrial training in reputed organizations to accrue industrial knowledge during the vacation period at the end of the second semester (Vacation period) for two weeks. The student has to find a suitable reputed industry related to their discipline (Public limited/Private Limited/ NGOs etc.) in consultation with the faculty in charge/Mentor and get approval from the head of the department and Departmental Committee

before going for an internship. After the internship, the student can submit the report to the departmental committee for assessment and awarding the mark/credit for it.

- **Format to be followed for Internship report (Two Copies)**

The format /certificate for internship report to be followed by the student are given below

- Title page -**Format of the title page**

Title of internship report

Internship report submitted in partial fulfilment of the requirement for the Master of Science in Microbiology to the Alagappa University, Karaikudi -630003.

By

(Student Name)

(Register Number)

University Logo

Department of Microbiology

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank-216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi - 630003

(Year)

- Certificate-(**Format of certificate – faculty in charge)**

This is to certify that the report entitled " ---------- " submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in Microbiology by Mr/Mis---------- (Reg No -------) under my supervision. This is based on the work carried out by him/her in the organization M/S----------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Research Supervisor

Date:_____

Certificate (HOD)

This is to certify that the Internship report entitled **" ---------- "** submitted by Mr/Mis**.----------(Reg No----------)** to the Alagappa University, in partial fulfilment for the award of the Master of Science in **-------** is a bonafide record of Internship report done under the supervision of----------,Dr.----------,Professor/ Associate Professor/Assistant Professor, Department of Microbiology , Alagappa University and the work carried out by him/her in the organization M/S ----------.

Place: Karaikudi

Date:_____

Head of the Department

Certificate-(**Format of certificate – Company supervisor or Head of the Organization)**

This is to certify that the Internship report entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in Microbiology by Mr/Mis---------- (Reg No:----------) under my supervision. This is based on the work carried out by him/her in our organization M/S ---------- for the period of three months or ------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Karaikudi

Date:_____

Supervisor or in charge

Declaration (student)

I hereby declare that the Internship Report entitled **" ---------- "** submitted to the Alagappa University for the award of the **Master of Science inMicrobiology** has been carried out by me under the supervision of---------- , Dr.----------, Professor/ Associate Professor/Assistant Professor, Department of Microbiology, Alagappa University, Karaikudi – 630 003. This is my original and independent work carried out by me in the organization M/S ---------- for the period of three months or---------- and has not previously formed the basis of the award of any degree, diploma, associateship, fellowship, or any other similar title of any University orInstitution.

Place: Karaikudi (----------)

Date:_____

- Acknowledgment
- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Organization profile /details	
4	Methods / Work	
5	Observation and knowledge gained	
6	Summary and outcome of the Internship study	
7	References	

- **No. of copies of the dissertation in final semester**

The candidate should prepare three dissertation documents and submit the same for the examiners' evaluation. After evaluation, one copy will be retained in the department library, one copy will be retained by the guide, and the student shall hold one copy.

F. Projects / Dissertation (Maximum Marks: 200**)**

- **Plan of work**

<u>Project/Dissertation</u>

The candidate shall undergo Project/Dissertation Work during the final semester. The candidate should prepare a scheme of work with the expertized guide in the field for the dissertation/project work. After completing the dissertation /project work, the candidate shall be allowed to submit it to the university departments at the end of the final semester.

- **Format to be followed for dissertation**

The format /certificate for the thesis to be followed by the student are given below

- Title page
- Certificate
- Acknowledgment
- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Review of literature	
4	Materials and methods	
5	Result	
6	Discussion	
7	Summary and Conclusion	
8	References	

- **Format of the title page**

Title of Dissertation

Dissertation/Project submitted in partial fulfillment of the requirement for the degree of Master of Science to the Alagappa University, Karaikudi -630003.

By

(Student Name)
(Register Number)
University Logo
Department of Microbiology

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank-216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi - 630003
(Year)

- **Format of certificates**

Certificate -Guide

This is to certify that the **Dissertation/Project** entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the degree of Master of Science in Microbiology by Mr/Mis ----------(Reg No---------) under my supervision. This is based on the results of studies carried out by him/her in the Department of----------, Alagappa University, Karaikudi-630 003. This dissertation/Project or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar titles or record of any University or Institution.

Place: Karaikudi

Date:_____ Research Supervisor

Certificate - (HOD)

This is to certify that the thesis entitled **" ---------- "** submitted by Mr/Mis ----------(Reg No: ----------) to the Alagappa University, in partial fulfilment for the award of the degree of Master of Science in Microbiology is a bonafide record of research work done under the supervision of Dr.----------, Professor/ Associate Professor/ Assistant Professor, Department of Microbiology, Alagappa University.

Place: Karaikudi Head of the Department

Date:

Declaration (student)

I hereby declare that the dissertation entitled **" ---------- "** submitted to the Alagappa University for the award of the degree of Master of Science in Microbiology has been carried out by me under the guidance of Dr.----------, Professor/ Associate Professor/ Assistant Professor, Department of , Alagappa University, Karaikudi – 630 003.

This is my original and independent work and has not previously formed the basis of the award of any degree, diploma, associateship, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi (----------)

Date:

14. Teaching methods

- Participation of students is essential; they are informed previously about the topic of the lecture.
- At the beginning of the lecture, the teacher inquiries about students' expectations and sets Objectives of the lecture.
- Some important points of the previous lecture are asked about.
- Students ask about non-clear points, and the teacher joins the previous with the new lecture.
- Teacher proposes some simple problems to be solved by students currently during the lecture.
- At the end, a summary of the content is presented by 2 or 3 students, followed by an organized summary by the teacher.

15. Attendance

Students must have earned 75% of attendance in each course to appear for the examination. Students who have earned 74% to 70% of attendance need to apply for condonation in the prescribed form with the prescribed fee. Students who have earned 69% to 60% of attendance need to apply for condonation in the prescribed form with the prescribed fee along with the Medical Certificate. Students who have below 60% of attendance are not eligible to appear for the End Semester Examination (ESE). They shall re-do the semester(s) after completion of the programme.

16. Examination

The examinations shall be conducted separately for theory and practicals to assess (remembering, understanding, applying, analyzing, evaluating, and creating) the knowledge required during the study. There shall be two systems of examinations, internal and external. The internal examinations shall be conducted as Continuous Internal Assessment tests I and II (CIA Test I & II).

A. Internal Assessment

The internal assessment shall comprise a maximum of 25 marks for each subject. The following procedure shall be followed for awarding internal marks.

Theory -25 marks

Sr.No	Content	Marks
1	Average marks of two CIA test	15
2	Attendance	2
3	Seminar/group discussion/quiz	4
4	Assignment/field trip report/case study report	4
	Total	25

Practical -25 Marks

1	Major Experiment	10 marks
2	Minor Experiment	5 marks
3	Spotter (2x 5/ 4 x4) or any other mode	10 marks
	Total	25 Marks

Project/Dissertation -50 Marks (assess by Guide and HOD)

1	Two presentations (mid-term)	30 Marks
2	Progress report	20 Marks
	Total	50 Marks

B. External Examination

- There shall be examinations at the end of each semester, for odd semesters in the month of October / November; for even semesters in April / May.
- A candidate who does not pass the examination in any course(s) may be permitted to appear in such failed course(s) in the subsequent examinations to be held in October / November or April / May. However, candidates who have arrears in Practical shall be permitted to take their Practical arrear examination only along with the Regular Practical examination in the respective semester.
- A candidate should get registered for the first-semester examination. Suppose registration is not possible owing to a shortage of attendance beyond condonation limit/regulation prescribed OR belated joining OR on medical grounds. In that case, the candidates are permitted tomove to the next semester. Such candidates shall re-do the missed semester after completion of the programme.
- For the Project Report/ Dissertation, the maximum marks will be 100 marks for project report evaluation, and for the Viva-Voce, it is 50 marks (if in some programmes, if the project is equivalent to more than one course, the project marks would be in proportion to the number of equivalent courses).
- Viva-Voce: Each candidate shall be required to appear for the Viva-Voce Examination (in defense of the Dissertation).

C. Scheme of External Examination (Question Paper Pattern)

Theory - Maximum 75 Marks

Section A	10 questions. All questions carry equal marks. (Objective type questions)	10 x 1 = 10 Marks	10 questions – 2 each from every unit
Section B	5 questions Either / or type like 1.a (or) b. All questions carry equal marks and each answer should not exceed one page or 250 words.	5 x 5 = 25	5 questions – 1 each from every unit
Section C	Essay type questions 5 out of 8 questions. All questions carry equal marks and each answer should not exceed two page	5 x8 = 40	5 question –Should cover all units

Practical –Maximum 75 Marks

Section A	Major experiment	15 Marks
Section B	Minor experiment	10 Marks
Section C	Experimental setup	5 Marks
Section D	Spotters (5 x 5 marks)	25 Marks
Section E	Record note	10 Marks
Section F	Vivo voce	10 Marks

17. Dissertation - Scheme of evaluation

Dissertation /Project report/Internship report	100 Marks
Vivo voce	50 Marks

18. Results

The results of all the examinations will be published through the Department where thestudent underwent the course as well as through University Website

19. Passing minimum

- A candidate shall be declared to have passed in each course if he/she secures not less than 40% marks in the End Semester Examinations and 40% marks in the Internal Assessment and not less than 50% in the aggregate, taking Continuous assessment and End Semester Examinations marks together.
- The candidates who do not obtain 50% in the Internal Assessment are permitted to improve their Internal Assessment marks in the subsequent semesters (2 chances will be given)by writing the CIA tests and by submitting assignments.
- Candidates, who have secured the pass marks in the End-Semester Examination and in the CIA but failed to secure the aggregate minimum pass mark (E.S.E + C I.A), are permitted to improve their Internal Assessment mark in the following semester and/or in University examinations.
- A candidate shall be declared to have passed in the Project / Dissertation / Internship if he /she gets not less than 40% in each of the Project / Dissertation / Internship Report and Viva-Voce and not less than 50% in the aggregate of both the marks for Project Report and Viva-Voce.
- A candidate who gets less than 50% in the Project / Dissertation / Internship Report must resubmit the thesis. Such candidates need to take the Viva-Voce again on the resubmitted Project report.

20. Grading of the Courses

The following table gives the marks, Grade points, Letter Grades and classifications meant to indicate the overall academic performance of the candidate.

Conversion of Marks to Grade Points and Letter Grade (Performance in Paper / Course)

RANGE OF MARKS	GRADE POINTS	LETTER GRADE	DESCRIPTION
90 - 100	**9.0 – 10.0**	**O**	**Outstanding**
80 - 89	**8.0 – 8.9**	**D+**	**Excellent**
75 - 79	**7.5 – 7.9**	**D**	**Distinction**
70 - 74	**7.0 – 7.4**	**A+**	**Very Good**
60 - 69	**6.0 – 6.9**	**A**	**Good**
50 - 59	**5.0 – 5.9**	**B**	**Average**
00 - 49	**0.0**	**U**	**Re-appear**
ABSENT	**0.0**	**AAA**	**ABSENT**

a) Successful candidates passing the examinations and earning GPA between 9.0 and 10.0 and marks from 90 – 100 shall be declared to have Outstanding (O).
b) Successful candidates passing the examinations and earning GPA between 8.0 and 8.9 and marks from 80 - 89 shall be declared to have Excellent (D+).
c) Successful candidates passing the examinations and earning GPA between 7.5 – 7.9 and marks from 75 - 79 shall be declared to have Distinction (D).
d) Successful candidates passing the examinations and earning GPA between 7.0 – 7.4 and marks from 70 - 74 shall be declared to have Very Good (A+).
e) Successful candidates passing the examinations and earning GPA between 6.0 – 6.9 and marks from 60 - 69 shall be declared to have Good (A).
f) Successful candidates passing the examinations and earning GPA between 5.0 – 5.9 and marks from 50 - 59 shall be declared to have Average (B).
g) Candidates earning GPA between 0.0 and marks from 00 - 49 shall be declared to have Re-appear (U).
h) Absence from an examination shall not be taken as an attempt.

From the second semester onwards the total performance within a semester and continuous performance starting from the first semester are indicated respectively **by Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA).** These two are calculated by the following formulate

GRADE POINT AVERAGE (GPA) = $\Sigma_i C_i \; G_i / \Sigma_i C_i$

$$GPA = \frac{\text{Sum of the multiplication of Grade Points by the credits of the courses}}{\text{Sum of the credits of the courses in a Semester}}$$

21. Classification of the final result

CGPA	Grade	Classification of Final Result
9.5 – 10.0 9.0 and above but below 9.5	**O+** **O**	First Class – Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	**D++** **D+** **D**	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	**A++** **A+** **A**	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	**B+** **B**	Second Class
0.0 and above but below 5.0	**U**	Re-appear

The final result of the candidate shall be based only on the CGPA earned by the candidate.

a) Successful candidates passing the examinations and earning CGPA between 9.5 and 10.0 shall be given Letter Grade (O+), those who earned CGPA between 9.0 and 9.4 shall be given Letter Grade (O) and declared to have First Class –Exemplary*.

b) Successful candidates passing the examinations and earning CGPA between 7.5 and 7.9 shall be given Letter Grade (D), those who earned CGPA between 8.0 and 8.4 shall be given Letter Grade (D+), those who earned CGPA between 8.5 and 8.9 shall be given Letter Grade (D++) and declared to have First Class with Distinction*.

c) Successful candidates passing the examinations and earning CGPA between 6.0 and 6.4 shall be given Letter Grade (A), those who earned CGPA between 6.5 and 6.9 shall be given Letter Grade (A+), those who earned CGPA between 7.0 and 7.4 shall be given Letter Grade (A++) and declared to have First Class.

d) Successful candidates passing the examinations and earning CGPA between 5.0 and 5.4 shall be given Letter Grade (B), those who earned CGPA between 5.5 and 5.9 shall be given Letter Grade (B+) and declared to have passed in Second Class.

i) Candidates those who earned CGPA between 0.0 and 4.9 shall be given Letter Grade (U) and declared to have Re-appear.

e) Absence from an examination shall not be taken as an attempt.

CUMULATIVE GRADE POINT AVERAGE (CGPA) = $\Sigma_n \Sigma_i C_{ni} \; G_{ni} / \Sigma_n \Sigma_i C_{ni}$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of Grade Points by the credits of the entire Programme}}{\text{Sum of the credits of the courses for the entire Programme}}$$

Where '**Ci**' is the Credit earned for Course i in any semester; '**Gi**' is the Grade Point obtained by the student for Course i and 'n' refers to the semester in which such courses were credited.

CGPA (Cumulative Grade Point Average) = Average Grade Point of all the Courses passed starting from the first semester to the current semester.

Note: * The candidates who have passed in the first appearance and within the prescribed Semesters of the PG Programme are alone eligible for this classification.

20. Maximum duration of the completion of the programme

The maximum period for completion of an **M.Sc., in Microbiology** shall not exceed Four semesters continuing from the first semester.

21. Conferment of the Master's Degree

A candidate shall be eligible for the conferment of the Degree only after he/ she has earned the minimum required credits for the Programme prescribed therefor (i.e. 90 credits). Programme).

22. Village Extension Programme

The Sivaganga and Ramnad districts are very backward districts where most people live in poverty. The rural mass is economically and educationally backward. Thus the aimof the introduction of this Village Extension Programme is to extend out to reachenvironmental awareness, social activities, hygiene, and health to the rural people of this region. The students in their third semester must visit any of the adopted villages within the jurisdiction of Alagappa University. They can arrange various programs to educate the rural mass in the following areas for three days based on the theme.1. Environmental awareness 2. Hygiene and Health. A minimum of two faculty members can accompany the students and guide them.

M.Sc Microbiology programme structure

S. No	Course Code	Title of the Course		T/P	Credits	Hours/ Week	Marks		
		I Semester					**I**	**E**	**Total**
1	**530101**	Core 1	General Microbiology	T	4	5	25	75	100
2	**530102**	Core 2	Microbial Biochemistry	T	4	5	25	75	100
3	**530103**	Core 3	Microbial Physiology	T	4	4	25	75	100
4	**530104**	Core 4	Lab-I: Lab in General Microbiology	P	4	6	25	75	100
6	**530105**	Core 5	Lab-II: Lab in Microbial Biochemistry and Microbial Physiology	P	4	6	25	75	100
7	**530501/ 530502**	DSE*-1 Biological techniques DSE*-1 Microbial Diversity and Taxonomy		T	3	3	25	75	100
		Library / Yoga/ counselling/Field trip		-		1			
		-		-	**23**	**30**	**150**	**450**	**600**
		II Semester							
8	**530201**	Core 6	Molecular Biology and Microbial Genetics	T	4	4	25	75	100
9	**530202**	Core 7	r DNA Technology	T	4	4	25	75	100
10	**530203**	Core 8	Food Microbiology	T	4	4	25	75	100
11	**530204**	Core 9	Lab III: Lab in Molecular Biology and Microbial Genetics	P	4	6	25	75	100
12	**530205**	Core 10	Lab-IV: Lab in r DNA Technology and Food Microbiology	P	4	6	25	75	100
13	**530503/ 530504**	DSE*2: Agriculture and Environmental Microbiology DSE*2: Microbial Ecology		T	3	3	25	75	100
14		Non-Major Elective **		T	2	3	25	75	100
15		Self-learning course (SLC) –MOOCs***		T	Extra credit - (EC)				
16		****Internship Program During vacation at the end of the second semester (Two weeks)		-	3	-	25	75	100
		-		-	**28**	**30**	**200**	**600**	**800**
		III Semester							
15	**530301**	Core 11	Medical Microbiology	T	4	4	25	75	100
16	**530302**	Core 12	Immunobiology	T	4	4	25	75	100
17	**530303**	Core 13	Industrial Microbiology	T	4	4	25	75	100
18	**530304**	Core 14	Lab V: Lab in Medical Microbiology	P	4	6	25	75	100
19	**530305**	Core 15	Lab-VI: Lab in Immunobiology and Industrial Microbiology	P	4	6	25	75	100
20	**530505/ 530506**	DSE*3: Algal Biotechnology DSE*3: Applied Microbiology I		T	3	3	25	75	100
21		Non-Major Elective **		T	2	3	25	75	100
22		Self-learning course (SLC) –MOOCs***		T	T				
		-		-	**25**	**30**	**175**	**525**	**700**
		IV Semester							
23	**530401**	Core 16	Applied Microbiology II	T	4	5	25	75	100
24	**530999**	Core 17	*****Dissertation Work	P	10	25	50	150	200
				-	**14**	**30**	**75**	**225**	**300**
			Total		**90 + EC**		**600**	**1800**	**2400**

DSE – Student Choice and it may be conducted by parallel sections.

** NME –Student have to select courses offered by other (Faculty) departments.

*** SLC- Voluntary basis

**** Internship report –Marks -Internal (25) by the institution where the student undergone the internship + External – 75 (Report (35) + Vivo-voce (40) = 75)

***** Dissertation report –Marks -Vivo-voce (50) + thesis (100) + internal (50) = 200

T-Theory

P-Practical

Non-Major Electives Course (NME) (For II Semester) - To be chosen by other PG degree students:

Subject	Contact Hrs / Week	Credits	Total No of Hrs Allotted	Max Marks I	Max Marks E	Total
Molecular Biology	3	2	30	25	75	100
Agriculture and Environmental Microbiology	3	2	30	25	75	100

Non-Major Electives Course (NME) (For III Semester) - To be chosen by other PG degree students:

Subject	Contact Hrs / Week	Credits	Total No of Hrs Allotted	Max Marks I	Max Marks E	Total
Medical Microbiology	3	2	30	25	75	100
Food and industrial Microbiology	3	2	30	25	75	100

Semester - I			
Course code: 530101	**General Microbiology**	**Credits:4**	**Hours: 5**
Objectives	➢ To inculcate knowledge on the fundamentals of microorganisms ➢ To learn the structural organization, morphology and reproduction of microbes ➢ To know the principles of Microscopy and advancements in Microscopy.		
Unit –I	**History and Scope of Microbiology** –Spontaneous generation theory, and the germ theory of disease– Contribution of Leuwenhoek, Louis Pasteur, Robert Koch, Edward Jenner, Joseph Lister, Winogradsky, Waksman and John Tyndall, Hargobind Khorana. Classification of microorganisms - Haeckel's three kingdom concept, Whittaker's five kingdom concept, Carl Woes' three-domain system, Principles and major characteristics-physiological, morphological and genetic characteristics used for microbial taxonomy. Bacterial classification (outline) according to Bergey's manual of systemic Bacteriology.		
Unit-II	**Ultrastructure of bacteria:** Morphological types, Cell wall of Gram negative, Gram positive bacteria, and halophiles. Cell wall synthesis and functions of the cell wall. Capsule composition and function. Cell membranes in Eubacteria, archaebacteria and cyanobacteria, Cell membrane functions. Periplasmic space. Structure and function of flagella, cilia and pili, gas vesicles, chlorosomes, carboxysomes, magnetosomes and phycobilisomes. Reserve food materials – polyhydroxybutyrate, polyphosphates, cyanophycin and sulphur inclusions. Bacterial endospores: Structure, biochemistry and genetics of sporulation. General account on Mycoplasma and Actinobacteria		
Unit III	**Algae, Fungi, Lichen, and Protozoa**: General characteristics, Classification, Structure and Reproduction of Algae: Chlorophyta (Green algae), Diatoms, Rhodophyta (Red algae), Fungi: Cell wall – chemical composition and functions, membranes and their functions, nutritional strategies of fungi. Structure and life cycle of fungi Ascomycetes (*Aspergillus*), Zygomycetes (*Mucor*), Basidiomycetes (*Agaricus*). Diversity and importance of lichen. Morphology and classification of Protozoa		
Unit IV	**Viruses:** Discovery, distinctive properties, morphology, and ultrastructure of Virus, Classification, Cultivation and Purification assay of the virus. Bacteriophages - structural organization and Classification, and life cycle - lytic, lysogenic. Viral-related agents - viroid and prion		
Unit V	**Types of media and culture preservation**: Types of growth media (natural, synthetic, complex, enriched and selective media). Anaerobic (thioglycolate, anaerobic chamber, Robertson's media, microaerophilic), liquid shake culture of aerobic bacteria Preservation and Maintenance of Microbial Cultures: Routine methods, liquid nitrogen preservation, freeze-drying (lyophilization), etc.		

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Outcomes	➢ Knowledge on historical perspectives of Microbiology ➢ Basic knowledge on different structures of microbes ➢ Ideas on the different types of microscope

Course Designed by: T. Kavitha

Semester – I			
Course code: 530102	**Microbial Biochemistry**	**Credits:4**	**Hours/week: 5**

Objectives	➢ To know the structural organization of bio-molecules ➢ To learn the characteristics of nucleic acids, enzymes and vitamins ➢ To acquire knowledge on secondary metabolites
Unit –I	**Carbohydrates:** Classification, Structure and properties of monosaccharides and disaccharides. Polysaccharides - starch, cellulose, agar- agar and peptidoglycan. **Metabolism and it's regulation:** Gluconeogenesis, glycolysis, kreb's cycle, pentose phosphate pathway or hexose monophosphate shunt, glyoxylate cycle and Entner Doudroff pathway.
Unit-II	**Amino acid and proteins:** Classification based on structure, polarity, biological importance and reactivity. Physical properties and chemical reactions. Biosynthesis and degradation of amino acids– an overall view. **Protein:** Classification, physical and chemical properties. Structure – Primary (peptide conformation, N- and C- terminal, peptide cleavage), Secondary (α-helix, sheet, random coil, Ramachandran plot), Tertiary and Quaternary structures of proteins.
Unit III	**Lipids and fatty acids:** Classification, structure, properties and functions.Phospholipid and cholesterol synthesis in *E.coli*. **Metabolism** - α, β and γ oxidation offatty acids and lipid peroxidation Nucleic acids: Structure, synthesis (de novo and salvage) and degradation of purines and pyrimidines.
Unit IV	**Enzymes and Vitamins:** Classification, chemical nature and properties. Factors affecting enzyme activity, Active site, **Enzyme inhibition**- Reversible, irreversible, allosteric inhibition, enzyme specificity, co-enzymes, **Mechanism of enzyme action**- Lock and key model, induced fit theory. Isozyme, ribozyme and abzyme. **Vitamins** – Properties of Vitamins. Vitamins as Co – factors and Co – enzymes.
Unit V	**Secondary Metabolites:** Antibiotics – Classification based upon mode of action. Biosynthesis and regulation of penicillin and streptomycin. **Microbial pigments** – Biosynthesis of Chlorophyll. **Microbial Toxins** –classification, structure and mode of actions,- *Salmonella* toxin, Cholera toxin, Botulism toxin and Aflatoxin.class

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Outcomes	➢ Knowledge on the metabolism of biomolecules ➢ General Information about nucleic acids, enzymes, and vitamins ➢ A clear idea on secondary metabolites and their biosynthetic pathways.

Course Designed by: Dr. T. Sathiamoorthi

Semester – I			
Course code: 530103	**Microbial Physiology**	**Credits:4**	**Hours: 4**
Objectives	➢ To develop a sufficient background for students about the growth of Microbes ➢ To learn about the microbial metabolism ➢ To acquire knowledge on microbial stress response		
Unit –I	**Growth of Bacteria:** Phases of growth. Growth kinetics - batch culture, continuous culture and synchronous culture - induction of synchrony. **Factors affecting growth** - nutrition, aeration, temperature and pH. **Physiological adaptation to extreme environmental conditions. Nutritional types and metabolic diversity** - types based on carbon, energy and electron sources.		
Unit-II	**Bacterial Photosynthesis:** Historical background. General types of microbial photosynthesis - oxygenic and anoxygenic. **Structure of photosynthetic pigments** – chlorophylls, bacteriochlorophyll, carotenoids and phycobilins. **Photosynthetic bacteria** - green sulfur and purple. **Mechanism of photosynthesis** - non-cyclic and cyclic electron transport. **Photo phosphorylation. Carbon assimilation** - Calvin, reverse citric acid cycle and hydroxyl propionate cycle.		
Unit III	**Nitrogen metabolism:** Nitrogen cycle - ammonification, nitrification, denitrification and nitrogen fixation. Nitrogenase enzyme, physiology of nitrogen fixation in symbiotic and free-living bacteria. Genetics of nitrogen fixation, acetylene reduction assay. Transamination and deamination.		
Unit IV	**Microbial stress responses** - osmotic stress and osmoregulation; aerobic to anaerobic transitions; oxidative stress; pH stress; acid tolerance; thermal stress, heat shock response; nutrient stress and starvation stress. **Fermentative pathways in specific groups of microbes:** alcoholic, lactic acid, formic, mixed, propionic, butyric, butanol, and butanediol fermentation. Anaerobic respiration.		
Unit V	**Bioenergetics:** Principles and laws of thermodynamics. **Coupling of chemical reactions** - TCA cycle, electron transport chain, and chemiosmotic theory of Mitchell. **Bio membranes:** Fluid mosaic model, transport across membrane - diffusion, osmosis, active transport and group translocation.		

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Subbarao, N.S. (2017). *Soil Microbiology.* Bengaluru: Medtec Publisher.

Outcomes	➢ Knowledge on the growth of Microbes ➢ General Information about the microbial metabolism ➢ A clear idea on energy production in the microbial cell.

Course Designed by: Dr. A. Arun

Semester – I			
Course code: 530104	Lab-I: Lab in General Microbiology	**Credits:4**	**Hours: 6**

Objectives	➢ To familiarize on basic microbiology techniques ➢ To learn the basic microbial biochemistry methods ➢ To know about the effect of environmental conditions on microbes

1. Principles and methods of sterilization
2. Preparation of media: nutrient broth, nutrient agar plate, soft agar.
3. Pure culture techniques: streak plate, spread plate and pour plate.
4. Motility determination – Hanging drop method and soft agar method
5. Isolation and enumeration of bacteria from different environmental samples.
6. Enumeration of bacteria - viable count (plate count) and total count (Haemocytometer count)
7. Direct microscopic observation of fungal spores and mycelium
8. Staining method: simple, negative, Gram's staining and spore staining.
9. Determination of microbial size by micrometry
10. Fungal slide culture KOH mount preparation
11. Measurement of growth rate and generation time by turbidometry method..

Reference and Textbooks:-

Aneja, K.R. (2003). *Experiments in Microbiology: Plant Pathology and Tissue Culture*, New Delhi: Wishwa Prakashan.

Aneja, K.N. (2018). *Lab Manual of Microbiology and Biotechnology*, Medtec Publisher

Cappuccino, J.H. and Sherman, N. (2014). *Microbiology – A Lab Manual (10th ed).* Singapore: The Benjamin Publishing Company.

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Trivedi, R. (2016). *Practical Mannual in Microbial Physiology and Industrial Microbiology*. New Delhi: SSDN Publishers

Outcomes	➢ Expertise in basic techniques of microbiology and biochemistry. ➢ Knowledge in the analysis and estimation of bio–molecules. ➢ Able to carry out microbial techniques.

Course Designed by: Dr. T. Kavitha

Semester – I			
Course code: 530105	Lab-II: Lab in Microbial Biochemistry and Microbial Physiology	**Credits:4**	**Hours: 6**

Objectives	➢ To familiarize on basic microbiology techniques ➢ To learn the basic microbial biochemistry methods ➢ To know about the effect of environmental conditions on microbes

1. pH metry - Preparation of buffer
2. UV visible spectrophotometry - Wavelength scan.
3. Chromatography
   - Paper chromatography – circular.
   - Thin layer chromatography - separation of amino acid.
4. Carbohydrates: Quantitative estimation of glucose and glycogen from bacterial and yeast cells.
5. Protein: Quantitative estimation of protein from bacterial yeast cells.
6. Enzyme: Estimation of alkaline phosphatase activity.
7. Environmental factor:
   - Effect of temperature on bacterial growth.
   - Effect of pH on bacterial growth.
8. Physiological groupings of bacteria.
   - Isolation of saccharophilic microorganisms (starch hydrolysis).
   - Proteolytic activity of microorganisms (casein and gelatin hydrolysis).
   - Lipolytic activity of microorganisms.
9. Utilization of Unusual compounds:
   - Microbial degradation of azodyes
10. Bioenergetics.
    - Cytochrome oxidase assay.
    - Catalase assay.
11. Nitrogen metabolism.
    - Nitrate reduction test.

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Aneja, K.R. (2003). *Experiments in Microbiology: Plant Pathology and Tissue Culture*, New Delhi:Wishwa Prakashan.

Aneja, K.N. (2018). *Lab Manual of Microbiology and Biotechnology*, Medtec Publisher Cappuccino, J.H. and

Sherman, N. (2014). *Microbiology – A Lab Manual (10th ed).* Singapore: The Benjamin Publishing Company.

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Trivedi, R. (2016). *Practical Mannual in Microbial Physiology and Industrial Microbiology.* New Delhi: SSDN Publishers.

Outcomes	➢ Expertise in basic techniques of microbiology and biochemistry. ➢ Knowledge in the analysis and estimation of bio–molecules. ➢ Able to carry out microbial techniques.

Course Designed by: Dr. T. Kavitha

Semester – I			
DSE-I: 530501	**Biological techniques**	**Credits:3**	**Hours: 3**
Objectives	➢ To inculcate the principles and applications of Microscopy ➢ To understand the working principles of chromatography ➢ To learn the concept and applications of physical-chemical techniques		
Unit I	**Microscopy and preparation techniques-** Principles and applications, simple, compound, dark, bright field, phase-contrast and fluorescent microscopes. Confocal laser scanning microscopy. Electron microscopy: SEM and TEM, Mechanism of image formation and contrast generation in SEM, Sample preparation methods for TEM		
Unit II	**Spectroscopy -** Electromagnetic spectrum, Beer Lambert's Law. UV/VIS Spectrophotometry, single beam, dual beam, Infrared spectroscopy, FTIR, Atomic absorption spectroscopy, Electron Spin Resonance Spectroscopy techniques, Spin label and H and C NMR spectroscopy. X-ray diffraction, Fluorescent spectroscopy, Quenching, principle, instrumentation of Mass spectrometry- MALDI-ToF MS, ESI-MS, ICP-MS.		
Unit III	**Centrifugation Techniques**: Principles, Swedberg unit, sedimentation coefficient, factors affecting sedimentation rate, clearing factor, rotors, their types and maintenance, determination of molecular weight by centrifugation, types of centrifuges, density gradient centrifugation, ultracentrifuges.		
Unit IV	**Chromatography and Electrophoresis -** Introduction and types of chromatography, paper, thin layer, gas and liquid, Rf value, Qualitative and preparative techniques, Gel permeation, ion exchange, HP-TLC, FPLC and affinity chromatography and instrumentation. Applications of Chromatographic Techniques in Microbiology. Electrophoresis - basic principles, PAGE - Native-PAGE, SDS-PAGE, Isoelectricfocussing and 2- Dimensional gels. Capillary electrophoresis. Principle and application of Agarose gel electrophoresis, DGGE, PFGE, Mobility shift electrophoresis		
Unit V	Bioethics and IPR: Definition of bioethics and ethical issues in biosciences, Ethical committee, Guidelines for research that involve animals, humans, microorganisms, Genetic engineering, Gene therapy, organ transplantation & Stem cells. Intellectual Property Rights (IPR): Tools of IPR – Patenting; Trademark; Trade secret; Copyrights; Geographical Indications; National and International Agencies (WTO, WIPO) involved in IPR and Patenting. Patenting of biological products.		

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Outcomes	➢ Students get knowledge on Microscopic techniques ➢ Understand the working principles and applications of spectroscopy and chromatographic techniques ➢ Get knowledge on physical chemistry

Course Designed by: Dr. A. Arun

Semester - I			
DSE -I: 530502	**Microbial Diversity and Taxonomy**	**Credits:3**	**Hours: 3**
Objectives	➢ To understand the ubiquitous nature of microbes ➢ To give basic knowledge on extremophiles ➢ To provide knowledge on the characteristics of Microbes.		
Unit -I	**Discovery of microbial world**: History, scope and relevance of microbiology. Current thoughts on microbial evolution, including the origin of life. Introduction to microbial biodiversity – distribution, abundance, ecological niche of bacterial, archaeal and Eukaryal. Prokaryotic diversity: The archaea – phylogenetic overview. Euryarcheota – extremely halophilic archaea, taxonomy and physiology of halophilic archaea. Methane producing archaea: methanogens – diversity and physiology. Thermo plasmatales–thermo plasma. Hyper thermophilic euryarchaeota: *Thermo coccales* and *Methano pyrus*.		
Unit-II	**Crenarcheota:** Habitat and energy metabolism, cold dwelling microbes (artic and antartic regions), hyperthermophiles – terrestrial, volcanic habitats –sulfo lobales and thermo proteales. Evolution and life at high temperature – heat stability of biomolecules, DNA stability, lipid stability. Limits to microbial existence.		
Unit III	**Systematics, occurrence, diversity, characteristic features and significance**: Spirochaetes - aerobic/microaerophilic motile, helical / vibriod - non-motile gram negative curved bacteria - gram negative aerobic rod and cocci - facultative anaerobic gram negative rod - anaerobic gram negative straight, curved & helical rods - sulfur reducing bacteria - anaerobic gram negative cocci - rickettsias and chlamydias – mycoplasmas - endosymbionts.		
Unit IV	**Diversity- characteristic features and significance**: Major Characteristics used in taxonomy. Gram positive cocci - endospore-forming; Gram positive rod and cocci-regular; non-sporing; gram positive rod – Irregular, non-sporing– Mycobacteria – *Nocardio formis*. Anoxygenic phototrophic bacteria, oxygenic photosynthetic bacteria, Nitrogen fixers, Nitrifying / Denitrifying bacteria. Aerobic chemo lithotrophic bacteria – budding and appendaged bacteria, sheathed bacteria, non-photosynthetic bacteria - Myxobacteria – Archea bacteria.		
Unit V	**Diversity, characteristic features and significance:** Nocardioform actinomycetes – actinomycetes with multilocular sporangia – actino planets – Streptomyces and related genera – Maduromycetes – Micromonospora - Thermonospora and related genera – Thermoactinomycetes – other genera. Fungi (general structure, nutrition and reproduction); **Evolutionary analysis:** distances, Cladistic and Phenetic methods. Sequence comparison, alignment and database searching – ClustalW, FASTA & BLAST. DNA barcoding.		

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Outcomes	➢ Students are able to differentiate various groups of Microbes ➢ Get knowledge on the adaptability of extremophiles ➢ Knowledge about microbial taxonomy.

Course Designed by: Dr. T. Kavitha

Semester - II			
Course code: 530201	**Molecular Biology and Microbial Genetics**	**Credits:4**	**Hours: 4**

Objectives	➢ To extend the knowledge on the structure and functions of genetic materials ➢ To focus on genome organization, transcription and translation process inprokaryotes ➢ To extend the knowledge on the molecular basis of mutation at the microbial level ➢ To focus on gene regulation and expression mechanisms ➢ To understand the principle role of plasmids and gene transfer methods
Unit –I	**Discovery of DNA**, **Structure of DNA** – A, B, and Z form. **Forms of DNA** – DNA heteroduplex, circular, superhelical DNA, twisted circle. **Properties of DNA** - denaturation, renaturation, melting curve, hyperchromicity. **Structure and types of RNA.** Structure of ribosomes. **Plasmids:** Types and Properties of plasmids – sex factors, drug resistant, colicinogenic, Agrobacterium Ti and broad host range plasmid. Copy number, replication- circular and theta. amplication and incompatibility.
Unit-II	**Origin of mutation. Biochemical basis of mutation:** Spontaneous mutation – random and non–adaptive mutation. Mutation rates. **Origin of spontaneous mutation** – isolation of mutants. **Selection of bacterial variation:** Direct - fluctuation test, indirect - replica plating. Mutagenesis and mutagenic agents. **Detection of mutagen** - Ames test, in *vitro* mutagenesis. Molecular basis of mutation. **DNA damage:** DNA damages, hit theory, UV radiation. **DNA repair:** post-irradiation effects on survival levels - photo reactivation, liquid holding recovery. **Biochemical repair mechanism:** excision, recombination and SOS repair.
Unit III	**Replication of DNA** - semi-conservative model, Meselson - Stahl experiment. **Enzymology of DNA replication** - DNA polymerase I, II and III; topoisomerase I and II; helicase; primase and gyrase. **Molecular mechanism of DNA replication**. Replication fork, origin and Okazaki fragments. **Transcription process in Prokaryotes:** Initiation - promotors, upstream and downstream sequences, transcription factors; **Elongation** - RNA polymerase, subunits; **Termination** - Rho-dependent and Rho-independent; nus A protein and antitermination. **Post-transcriptional processes**: RNA processing, rRNA and tRNA processing, RNA Editing: Post-transcriptional gene regulation.
Unit IV	**Genetic code:** Elucidation of triplet code, code characteristics and codon dictionary. Reading frames, sense and nonsense code. **Degeneracy -** wobble hypothesis, the universality of genetic code. **Process of translation in prokaryotes:** Initiation and Termination. Role of rRNA in protein synthesis. **Post-translational modifications** - Protein modification, folding, chaperones, transportation; signal hypothesis, protein degradation. **Gene concept** - regulation of bacterial gene expression. **Lactose system** - coordinate regulation, Lac components, positive and negative regulation, catabolite repression. **Tryptophan operon** – regulation and attenuation. **Arabinose operon and regulation.**
Unit V	**Gene transfer and recombination: Transformation. Conjugation. Transduction:** DNA generalized and specialized transduction, **Recombination:** Types - homologous or general, site specific and random recombination, general recombination between homologous DNA- Holliday model, double strand model of general recombination, enzymes involved in recombination rec - proteins.

Reference and Textbooks:-

Benjamin Lewin. (2007). *Genes XI.* New York: Oxford University Press.

Cummings, M.R., Klug, W.S. (1995). *Essentials of Genetics (9th ed).* Pearson Publisher.

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Sambamurty, A. V. S. S. (2007). *Molecular Genetics*. Narosa Publication.

Sanders, M.F. and Bowman, J.L. (2018*). Genetic Analysis: An Integrated Approach.* Pearson Publisher. Stanley R. Maloy, John E.C. and Freifelder, D. (2008). *Microbial Genetics.* New Delhi: Narosa Publishing House. Stryer, L. (2019). *Biochemistry (9th ed).* New York: W.H. Freeman and Company. Watson, J.D., Hopkins, N.H., Roberts, J.W., Steitz, J.A. and Weiner, A. M. (2013). *Molecular Biology of the Gene (17th ed).* Tokyo: The Benjamin Cummings Publishing Company Inc.	
Outcomes	➢ Receive elaborate knowledge on mutation ➢ A better understanding of gene regulation ➢ Get thorough knowledge on gene transfer mechanisms in microbes.

Course Designed by: Dr. A. Arun

Semester – II			
Course code: 530202	**r DNA technology**	**Credits:4**	**Hours: 4**
Objectives	➢ To learn about the various enzymes involved in r DNA Technology ➢ To know the principles of c DNA construction and amplification methods. ➢ Making aware of synthesis of recombinant products		
Unit -I	**DNA modifying enzymes:** Polymerases, restriction endonucleases, polynucleotide kinase, terminal deoxynucleotidyl transferase, DNase, Methylase, phosphatases, ligases. Cloning vectors – plasmids, cosmids, phasmids, phagemids, expression vectors, plasmid vectors – $p^{BR}322$ and $p^{UC}18$, integrating shuttle vector –YAC vectors, viral vector – SV 40 and adenovirus. Lac Z promoter – expression system – Lambda, PL /PR Promoter, T^7 promoter, Sp6 promoter, SV – 40 promoter, CaMV 35s promoter.		
Unit-II	**Cloning methodologies:** α complementation, sticky and blunt-end cloning. Cloning from mRNA – synthesis of cDNA, cloning cDNA– cDNA library. Cloning from genomic DNA – genomic library. Shotgun cloning. Screening of recombinant – phenotypic expression of characters – Blotting techniques – western, northern and southern. Mapping of human genes – Human genome project.		
Unit III	**PCR:** gene amplification, primer designing, optimization, variation in the PCR (RAPD, RFLP, RACE, RT-PCR) DNA sequencing – Sanger – Coulsen's method, Maxam Gilbert's method automated sequencing and microarray.		
Unit IV	**Cloning:** human insulin, interferon in *E.coli.* Recombinant vaccine development – HBs Ag in yeast. Cloning for commercial production of antibiotics (Penicillin). Bio steroid transformation. Production of biopolymers – Xanthumgum. Melanin biosynthesis in *E.coli*, adhesive biopolymer in yeast.		
Unit V	**Gene silencing and antisense technology:** Types and mechanism of gene silencing. Genetic factors of silencing, formation of antisense mRNA, inhibition of gene expression by antisense RNA. Gene silencing in crop plants: tomato. Si RNA and disease control. Plant genetic engineering: Ti plasmid, CaMV vector, Direct DNA delivery methods – microprojectile bombardment, microinjection, electroporation. Liposome-mediated gene transfer and DNA/calcium phosphate co-precipitate method. Gene therapy		

Reference and Textbooks:-

Brown, T.A. (2006). *Gene Cloning and DNA Analysis: An Introduction (5th ed).* Oxford: Blackwell Publishing.

Glick, B.K. and Pasternak, J.J. (2010). *Molecular Biotechnology*: *Principles and Applications of Recombinat DNA (4th ed).* Washington: ASM Press.

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Stryer, L. (2019). *Biochemistry* (9th ed). New York: W.H. Freeman and Company.

Susan, R.B.(2008). *Biotechnology*, New Delhi: Cengage Learning Pvt. Ltd.

Thieman, W.J. and Palladino, M.A. (2009). *Introduction to Biotechnology*, Noida: Dorling Kindersley India Pvt. Ltd.

Veer Bala Rastogi, (2016). *Principles of molecular biology*. Medtech Publishers

Watson, J.D., Hopkins, N.H., Roberts, J.W., Steitz, J.A. and Weiner, A. M. (2013). *Molecular Biology of the Gene* (7th ed). Tokyo: The Benjamin Cummings Publishing Company Inc.

Outcomes	➢ Students come out with basic ideas on cloning vehicle ➢ Enable them to know about c DNA and amplification products. ➢ Familiar with the construction of recombinant DNA.

Course Designed by: Dr. T. Kavitha

Semester - II			
Course code: 530203	**Food Microbiology**	**Credits:4**	**Hours/week: 4**

Objectives	➢ To learn about food spoilage factors ➢ To know the preservation methods ➢ To make aware of food-borne disease
Unit -I	**History and development of Food microbiology** Role and Significance of Microorganisms in Foods, Outline of food spoilage and preservation, **Food safety and microbiological Quality Assurance**. Micro-organisms and Food Materials-Diversity of Habitat, **Micro-organisms in the Atmosphere** - Airborne Bacteria, Airborne Fungi, Micro-organisms of Soil, **Micro-organisms of Water**, Micro-organisms of Plants, Micro-organisms of Animal Origin.
Unit-II	**Factors Affecting the Growth and Survival of Micro-organisms in Foods**. - Microbial Growth, Intrinsic Factors- Nutrient Content, pH and Buffering Capacity, Redox Potential, Antimicrobial Barriers and Constituents, **Water Activity and Extrinsic Factors** - Relative Humidity, Temperature and Gaseous Atmosphere
Unit III	**Microbiology of Food Preservation** - Heat Processing, Irradiation, High-pressure Processing – Pascalization, Low-temperature Storage and Chemical Preservatives. Aseptic Packaging, Mano-thermo-sonication, Microbiologicalquality standards of food, FDA, HACCP, ISI, AGMARK.
Unit IV	**Fermented dairy products:** Microbes associated with raw milk: Significance of specific groups of microorganisms in milk - psychrotrophic, mesophilic, thermoduric and thermophilic bacteria – their morphological and biochemical characteristics and classification. Microbial contaminations in raw milk, and their sources during various stages of production – milking, chilling, storage and transportation with special reference to psychrotrophic microorganisms. Fermented dairy products – microbes fermentation of butter milk, cream, yogurt, kafir, kumiss, acidophilous milk and cheese production and its types.
Unit V	**Food spoilage, preservation and foodborne diseases:** organism involved in spoilage of fruits and vegetables, cereal and cereal products, meat and meat products. Foodborne diseases- Bacterial foodborne diseases- (*Staphylococcal* intoxication, Botulism, Salmonellosis, Shigellosis, EPEC Diarrhoea,; Food-borne fungi- Mycotoxins- Aflatoxicosis, Deoxynivalenol, Mycotoxicosis, Ergotism. Food Borne Viral Pathogens- (Norwalk virus, Reovirus, Rotavirus, Adenovirus, Parvovirus, Hepatitis A Virus)

Reference and Textbooks:-

Adams, M.R. and Moss, M.O. (2008). *Food Microbiology*. UK: RSC Publishing, Cambridge.

Aneja, K.N. (2018). *Modern Food Microbiology*, Medtec Publisher.

Bhatnagar, R. (2017). *Food Microbiology*, Crescent Publishing Corporation.

Blackburn C. de W. (2006), *Food spoilage microorganisms*. UK: Woodhead Publishing, Cambridge

Cruger, W. and Crueger, A. 1995, Biotechnology. Black Well Scientific Publications, Oxford.

Deak, T. and Beuchat, L.R. (1996). *Hand Book of Food Spoilage yeasts*, CRC

Dick M, (2017). *Food Microbiology An Introduction* (2nd ed). Bengaluru: Medtech.

Frazier WC and Westhoff DC. (2014) Food microbiology, TATA McGraw Hill Publishing Company Ltd. 5th edition, New Delhi.

Jay J.M. (2000). *Modern Food Microbiology* (6th ed). New York: Chapman & Hall.

Foster W. M. *Food Microbiology*, CBS Publication

Prescott, L.M., Harley, J.P. and Helin, D.A. (2008). *Microbiology* (5th ed). New York: McGraw Hill.

Outcomes	➢ A better understanding of the cause of microbes in food spoilage ➢ Get information regarding food preservation techniques ➢ Enable them to work in food fermentation industries

Course Designed by: Dr. T. Sathiamoorthi

Semester - II			
Course code: 530204	Lab-III: Lab in Molecular Biology and Microbial Genetics	**Credits: 4**	**Hours: 6**

Objectives	➢ To impart knowledge on the isolation and estimation of nucleic acids ➢ To practice the students in rDNA technology ➢ To make them experts in food Microbiology

1. Isolation of genomic DNA from bacteria.
2. Isolation of genomic DNA from yeast.
3. Isolation of plasmid DNA from bacteria.
4. Extraction of total RNA from bacteria and yeast
5. Estimation of nucleic acids
   a) UV - VIS spectrophotometer analysis.
   b) Analysis of nucleic acids by agarose gel electrophoresis.
6. Detection of proteins by SDS-PAGE.
7. Determination of percentage of the killing of bacterial cells by UV rays.
8. Plotting of UV survival curve.
9. Reversion of auxotrophy.

Reference and Textbooks:-

Aneja, K.R. (2003). *Experiments in Microbiology: Plant Pathology and Tissue Culture.* New Delhi: Wishwa Prakashan.

Ausubel, F.M., Roger, B., Robert E. Kingston, David A. Moore, Seidman J.G., John A. Smith. and Kelvin, S. (1992). *Short Protocols in Molecular Biology* (3rd ed). New York: Jolm Wiley &Sons Inc.

Berger, S.L. and Kimmel, R. (1987). *Guide to Molecular Cloning Techniques*. New York: Academic Press, Inc.

Brown, T.A. (1998). *Molecular Biology Lab Fax 11 Gene Analysis*. London: Academic Press. Cappuccino, J.H. and Sherman, N. (2007). *Microbiology – A Lab Manual* (7th ed). Singapore: The Benjamin Publishing Company.

Malov, S.R. (1990). *Experimental Techniques in Bacterial Genetics*. Boston: Jones and Bartlett Publishers.

Miller, J.H. (1992). *A Short Course in Bacterial Genetics: A Lab Manual &Hand Book for E. coli and related Bacteria*. Cold Spring Harbour: Cold spring Harbor Lab press.

Palanivel, P. (2000). *Laboratory Manual for Analytical Biochemistry & Separation Techniques.* School of Biotechnology, Madurai Kamaraj University, Madurai.

Sambrook, I., Fritsch, E.F. and Maniatis, T. (2001). *Molecular Cloning 1, 2, 3 - A Laboratory Manual* (3rd ed). USA: Cold Spring Laboratory Press.

Verma, A.S., Surajit, D and Anchal, S. (2014). *Laboratory Manual for Biotechnology.* New Delhi: S. Chand and Company Ltd.

Outcomes	➢ Trained in isolation of nucleic acids ➢ Become familiar in rDNA technology ➢ Expertise in food Microbiology

Course Designed by: Dr. A. Arun

Semester - II			
Course code: 530205	Lab-IV: Lab in r DNA Technology and Food Microbiology	**Credits: 4**	**Hours: 6**

Objectives	➢ To impart knowledge on the isolation and estimation of nucleic acids ➢ To practice the students in rDNA technology ➢ To make them experts in food Microbiology

1. Isolation of streptomycin-resistant mutants using gradient plate technique.
2. AMES test.
3. Isolation of auxotrophic mutant.
4. Uninterrupted bacterial conjugation.
5. Isolation of phage from the septic tank.
6. P1 Transduction.
7. Isolation of p^{BR} 322/ p blue script by alkaline detergent method - A miniprep procedure
8. Recovery of DNA from gels.
9. Acrylamide gel electrophoresis and silver staining of the digested plasmid.
10. Cloning of DNA fragment in p^{BR}322/pbluescript – insertion inactivation/ blue white selection.
11. Viable count of bacteria in milk.
12. Methylene Blue Dye reduction test.
13. Resazurin dye reduction test.
14. Phosphatase test.
15. Litmus milk reaction
16. Potability analysis of drinking water.

Reference and Textbooks:-

Aneja, K.R. (2003). *Experiments in Microbiology: Plant Pathology and Tissue Culture.* New Delhi: Wishwa Prakashan.

Ausubel, F.M., Roger, B., Robert E. Kingston, David A. Moore, Seidman J.G., John A. Smith. and Kelvin, S. (1992). *Short Protocols in Molecular Biology* (3rd ed). New York: Jolm Wiley &Sons Inc.

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Cappuccino, J.H. and Sherman, N. (2007). *Microbiology – A Lab Manual* (7th ed). Singapore: The Benjamin Publishing Company.

Malov, S.R. (1990). *Experimental Techniques in Bacterial Genetics*. Boston: Jones and Bartlett Publishers.

Miller, J.H. (1992). *A Short Course in Bacterial Genetics: A Lab Manual &Hand Book for E. coli and related Bacteria*. Cold Spring Harbour: Cold spring Harbor Lab press.

Palanivel, P. (2000). *Laboratory Manual for Analytical Biochemistry & Separation Techniques.* School of Biotechnology, Madurai Kamaraj University, Madurai.

Sambrook, I., Fritsch, E.F. and Maniatis, T. (2001). *Molecular Cloning 1, 2, 3 - A Laboratory Manual* (3rd ed). USA: Cold Spring Laboratory Press.

Verma, A.S., Surajit, D and Anchal, S. (2014). *Laboratory Manual for Biotechnology.* New Delhi: S. Chand and Company Ltd.

Outcomes	➢ Trained in isolation of nucleic acids ➢ Become familiar with rDNA technology ➢ Expertise in food Microbiology

Course Designed by: Dr. A. Arun

Semester - II			
DSE -II:530503	**Agriculture and Environmental Microbiology**	**Credits:3**	**Hours/week:3**

Objectives	➢ To create awareness on soil Microbiology ➢ To give knowledge on plant pathogen interaction and its control ➢ To inculcate on environmental microbiology
Unit -I	**Diversity and distribution of microorganisms in soil;** Soil Microflora- Bacteria, Fungi and Actinomycetes. Classification, physical, chemical properties and structure of soil. **Microbial interactions** - mutualism, synergism, commensalism, amensalism, parasitism, predation and competition. **Microbial interactions with plants**–phyllosphere, mycorrhizae, rhizosphere and symbiotic association in root nodules. **Biofertilizer** – VAM, *Rhizobium, Frankia, Azospirillum*, *Azotobacter*, Cyanobacteria, Phospho bacteria and *Azolla*.
Unit-II	**Bacterial diseases of agricultural crops** - pathogens, symptoms, control measures with reference to paddy, cotton, maize, tomato, citrus, mango and potato. **Plant protection** – Phenolics – phytoalexins and related compounds. Bioinsecticides – viral (Baculovirus, NPV)- bacterial (*Bacillus thuringiensis*) andfungal (*Trichoderma*) - a brief note.
Unit III	**Bio-geo chemical cycles in soil** – Carbon cycle, Nitrogen cycle – Nitrogen fixation, nitrification, denitrification, sulfur, iron and phosphorus cycles. **Aerobiology** – a brief introduction - droplet nuclei – aerosols - air-borne transmission of microbes and diseases and assessment of air quality.
Unit IV	**Aquatic microbiology** - factors affecting microbial growth – temperature – pressure – light – salinity - turbidity – pH -inorganic and organic constituents. **Aquatic habitats** - freshwater - lakes, ponds and streams; **marine habitats** - estuaries, deep sea, hydrothermal vents, saltpans, coral reefs and mangroves and their microbial communities; **zonation** – food chain and food web.
Unit V	**Types of wastes** - solid and liquid wastes. **Treatment of solid wastes** - Thermal Treatment: Incineration, Gasification, Pyrolysis and Open Burning- **Dumps and Landfills**: Sanitary landfills, Controlled dumps, Bioreactor Landfills-**Biological Waste Treatment**: Composting, Vermicomposting and termi composting. **Treatment of liquid wastes** –primary, secondary, tertiary treatment; anaerobic (methanogenesis), aerobic, Trickling, activated sludge, oxidation pond. Production of biogas from waste.

Reference and Textbooks:-

Alexander M. (1997). *Introduction to soil microbiology*, New York: John Wiley & Sons, Inc.

EcEldowney S., Hardman, D.J. and Waite, S. (1993). *Pollution Ecology and Biotreatment*. Longman Scientific Technical.

Grant, W.D. and Long, P.L. (1981). *Environmental Microbiology*. Blalckie Glasgow and London.

Madigan, M.T., Martinka, M., Parker, J. and Brock, T.D. (2000). Twelth Edition, *Biology Microorganisms*, New Jerry: Prentice Hall.

Mark Wheelis, (2010). *Principles of Modern Microbiology*, New Delhi: Jones & Bartlett India Pvt. Ltd.

Mehrotra, R.S. (1983). *Plant Pathology*, New Delhi: Tata McGraw Hill Publishing Company Ltd.

Pandy, B.P. (1997). *Plant Pathology (Pathogen & Plant Disease)*, New Delhi: S.Chand& Company Ltd.

Ray Chadhuri, S.P. (1977). *A Manual of Virus Diseases of Tropical Plants*, New Delhi: MacMillan Company of India Ltd.

Rengaswami, G. and Rajagopalan, S. (1973). *Bacterial Plant Pathology*. Coimbatore: Tamil Nadu Agriculture University.

SubbaRao, N.S. (1995). *Soil Microorganisms and Plant Growth* (3rd ed). New Delhi: Oxford & IBH Publishing Co. Pvt. Ltd.

Outcomes	➢ Acquire knowledge on soil microbiology ➢ Understand the biogeochemical cycles prevail in the environment. ➢ Able to know about principles and techniques in waste treatment.

Course Designed by: Dr. T. Sathiamoorthi

Semester - II			
DSE -II: 530504	**Microbial Ecology**	**Credits:3**	**Hours: 3**
Objectives	➢ To create awareness on the evolutionary relationship of ecosystem ➢ To learn about the individual ecosystem and its interactions. ➢ To understand the concepts of community ecology		
Unit -I	**Introduction microbial ecology**:- overview, history and scope of microbial ecology. Population ecology: properties of population–density, natality and fecundity, mortality, longevity and senescence, immigration and emigration. Microbial population ecology – population growth, density dependence and independence, r and k selection. Species concept – universal, biological, phenetical, evolutionary and phylogenetic. Speciation – bacterial, mismatch repair as a speciation mechanism, rapid speciation, operons, genome economization and speciation, hypermutation, genome reduction.		
Unit-II	**Concepts of microbial ecology**:- definition and terminology, Ecology of individuals- the study of individual microorganisms, genetic individuals, ramets. ecological individual, niche. Abiotic constraints (temperature, pH, nutrient source, electron acceptor, redox, pressure, and light), metapopulation, dispersal, modularity, source and sinks, and population ecology of genes. Source of phenotypic and genotypic variation, gene ecology.		
Unit III	**Population and spatial stability**:- Uniformity of populations, adaptation, the population in time. Bacterial communication: Quorum sensing – the evolutionary implication of quorum sensing, cell-cell communication in bacteria, quorum sensing and evolution, disruption or manipulation of quorum sensing response, oligotrophic state of nature, starvation survival, aging, senescence, death, dormancy or resting state and miniaturization.		
Unit IV	**Microbial Interactions**:- with their physical and chemical environment, Species interaction and processes: Species interaction, proliferation hypothesis. Interactions with the biotic environment: symbiosis, competition, parasitism, predation, Negative relationship – parasitism, predation, bacterial and viral interaction, microbial loop and bacteria as predators. Neutral relationship: positive relationship - metabiosis and symbiosis.		
Unit V	**Community ecology**:-Water communities - hydrosphere ecology of fresh water, composition and activity of freshwater microbial communities, physical and chemical factors, estuaries and marine water environment; characteristics and stratification of the ocean, composition and activity of marine microbial communities, the role of microbes in the aquatic environment and lithosphere. Soil communities - introduction to soil formation, rock and minerals, soil horizon, soil texture, organic matter, chemical properties of soil, and soil microbial communities. Biofilm communities, phylogenetics, andcommunity ecology.		

Reference and Textbooks:-(APA Format)

Atlas, R.A. and Bartha, R. (2000). *Microbial Ecology, Fundamentals and Application*. New York: Benjamin Cummings.

Dubey, R.C. and Maheswari, D.K. (2013*). A text book of Microbiology* (Revised). New Delhi: S. Chand and Company Ltd.

Madigan, M.T., Martinka, M., Parker, J. and Brock, T.D. (2000). *Biology Microorganisms* (12th ed)., New Jerry: Prentice Hall.

Pelczar, M.J., Schan, E.C. and Kreig, N.R. (2010). *Microbiology – An application based approach* (5th ed). New Delhi: Tata McGraw Hill Publishing Company Limited.

Prescott, L.M., Harley, J.P. and Helin, D.A. (2008). *Microbiology* (9thed). New York: McGraw Hill.

Saha, T.K. (2010). *Ecology and Environmental Biology*. Kolkata: Books and Allied Pvt. Ltd.

Schlegal, H.G. (1995). *General Microbiology* (7thed). Cambridge: Cambridge Univeristy Press.

Stanier, R., Lingraham, Y., Wheelis, M.L. and Painter, R.P. (1986). *General Microbiology* (5thed). London: Macmillan.

Tortora G.J., Funke, B.R. and Case, C.L. (2009). *Microbiology* (9thed). Noida: Dorling Kindersely (India) Pvt. Ltd.

VaunMc Arthur (2009) *Microbial Ecology – An Evolutionary approach*, Elsevier publications, Academic Press.

Course Designed by: Dr. V. Balasubramanian

Semester - III			
Course code:530301	**Medical Microbiology**	**Credits: 4**	**Hours/week: 4**

Objectives	➢ To inculcate on the role of normal flora and pathogenic microbes ➢ To understand the pathogenesis of various diseases ➢ To understand the various clinical microbiological techniques.
Unit -I	**Laboratory management** – Safety in containment laboratory. Rules and regulationsto be followed in a microbiology laboratory. Collection, transport of clinical samplesand laboratory waste disposal system. Microbiological examination of urine, blood, feces, cerebrospinal fluid, throat swabs, sputum, pus and wound exudates. **Normal flora of human systems** – skin, respiratory tract, gastrointestinal tract and genitourinary tract. Nosocomial infections. Storage method of clinical samples in laboratory.
Unit-II	Morphology, classification, cultural characteristics, pathogenicity, laboratory diagnosis and prevention of infections caused by the following organisms: **Gram positive cocci**– *Staphylococci, Streptococci.* **Gram negative cocci**– *Gonococci.* **Gram positive non-spore-forming bacilli**: aerobic – *Corynebacteria* and anaerobic-*Actinomyces*. **Gram positive spore forming bacilli**: aerobic- *Bacillus anthracis* and anaerobic *Clostridia.Corynebacterium diptheriae.*
Unit III	General characters, pathogenesis, laboratory diagnosis and control measures of:**Gram negative non-spore-forming bacilli**: Aerobic- *Bordetella*. Small gram negative facultative anaerobic bacteria – *Yersinia*. **Enteric gram negative bacilli** – *Vibrio*, *E.coli* and *Salmonella.* **Acid fast bacteria** – *M. tuberculosis, M.* leprae.Cell wall less bacteria – *Mycoplasma*. **Spirochaetes** –Leptospirosis. Sexually transmitted diseases.
Unit IV	General characteristics, pathogenesis and laboratory diagnosis and control measures of **Yeast**–*Cryptococcus neoformans.* **Yeast-like fungus** – *Candida spp*. **Filamentous fungi** – *Aspergillus* and *Penicillium.* Dimorphic fungi, yeast morphology, general characteristics and reproduction. – *Blastomyces dermatitis.* Classification, structure and reproduction of fungi, general characteristics of Intracellular parasites– *Cryptosporidium* and *Plasmodium.* Intralumen parasites – *Entameoba histolytica* and *Ascaris lumbricoides.* Parasitic zoonosis– *Toxoplasma* and *Taenia*
Unit V	Morphology, pathogenesis and laboratory diagnosis and control measures of: **DNA viruses** –Hepatitis B virus. **RNA viruses** – Flavi virus (dengue), Retrovirus – HIV. **Viral zoonosis** -rabies. **Classification of antibiotics based on the mode of action**: antibacterial (Penicillin), antiviral (Amantadine), antifungal (Amphotericin), antiparasitic drugs (Quinine and Metronidazole). Infectious diseases- Definition of emerging & re-emerging diseases. Factors contributing to emergence. Examples (Chickungunya, Zika virus, H1N1 and Ebola). National programs in the prevention of infectious diseases. Papilomaviridae- Human papilloma viruses and Rhabdoviridae

Reference and Textbooks:-

Anathanarayan R and Jeyaram Panikers C.K. (2013). *Text Book of Microbiology* (9th ed). New Delhi: Jain book depot.

Arora D.R., Brij Bala Arora.(2015). *Textbook of Microbiology*. Chennai: CBS.

Awetz Melnick and Adelberg's. (2010). *Medical Microbiology, 21st Century.* Appleton & Lange.

Bhattacharjee R.N.(2015). *Introduction to Microbiology* (1st ed). New Delhi: Kalyani Publishers.

Connie R Mahon. (2010). *Textbook of Diagnostic Microbiology* (3rd ed). Pearson.

David Greenwood, Richard Slack, John Peutherer. (2012). *Medical Microbiology.* Churchill Livingstone.

Jesse Russell, Ronald Cohn. (2012). *Medical Microbiology.* Book on Demand Ltd.

Myra Wilkinson. (2011). *Medical Microbiology.* Scion Publishing Ltd

Patrick R. Murray. (2015). *Medical Microbiology.* Elsevier

Patrick Murray & Ken Rosenthal & Michael Pfalle. (2015). *Medical Microbiology* (8th ed). New York: Academic Press.

Outcomes	➢ Get information about various mechanisms of infection ➢ Knowledge on clinical lab techniques

	➢ Acquire knowledge on control measures of diseases

Course Designed by: Dr. T. Sathiamoorthi

Semester - III			
Course code: 530302	**Immunobiology**	**Credits:4**	**Hours: 4**
Objectives	➢ To provide knowledge on the human immunity system ➢ To understand the mechanism of antigen-antibody reaction ➢ To inculcate the principles of vaccine development.		
Unit -I	**History and scope of immunology**: Types of immunity: innate-components-physical, physiological defenses: complement, acute phase proteins and adaptive immunity, Acquired immunity: (specific) natural, artificial, active and passive immunity. Inflammatory response; Phagocytic system- mononuclear phagocytes, macrophages, neutrophils, Natural killer cells, mast cells, basophils, and eosinophils. Physiology of immune response – humoral and cell-mediated immunity. Lymphoid organs – primary and secondary. Barriers of the immune system- Haematopoietic stem cells. Transfusion, rh incompatibility		
Unit-II	**Antigens**: characteristics, types, cross-reactivity, hapten, adjuvant, immunogenicity and antigenicity. Immunoglobulin – types structure and functions. Engineered antibodies. Antigen- Antibody interactions. Immuno technology – hybridoma and monoclonal antibodies. Super antigen, antigen processing and presentation to T-lymphocytes.		
Unit III	**Immune effector mechanisms:** Cytokines – properties and functions. Complement components – classical and alternate pathways, complement activation, and complement deficiencies. Hypersensitivity – anaphylaxis, cytotoxic, immune complex deposition and cell- mediated. Autoimmunity – idiotype, network and autoimmune diseases. Mechanism of immune regulation – tolerance. DTH reponse		
Unit IV	**Immunity to infectious diseases:** bacterial (*Mycobacterium tuberculosis*), viral (HIV), protozoan (*Leishmania*). Vaccines: Types – inactivated, subunit, synthetic, DNA and live attenuated vaccines- Immunoinformation, immunomodulation in infection.		
Unit V	**Transplantation immunology:** Graft versus host reactions. Structure, functions of class I and class II MHC molecules, HLA typing. Principles of tumor immunology: Immunodiagnosis based on antigen and antibody interaction - precipitation, agglutination, EIA, RIA, Immunodiffusion, Immunoelectrophoresis, Immunofluorescence, Immunoprecipitation, flow cytometry and immunofluorescence, ELI spot techniques, cancer immunology		

Reference and Textbooks:-

Abul, K. Abbas Andrew H. H. Lichtman& Shiv Pillai. (2015). *Basic Immunology, Functions and Disorders of the Immune System* (5th ed). Elsevier.

Abul K. Abbas & Andrew H. Lichtman & Shiv Pillai. (2014). *Cellular and Molecular Immunology* (8th ed). Elsevier.

Barbara, J. A., Regan, F. A., & Contreras, M. (Eds.). (2008). *Transfusion microbiology*. Cambridge University Press.

Coico, R., & Sunshine, G. (2015). Immunology: a short course. John Wiley & Sons.

Day, M. J., & Schultz, R. D. (2014). *Veterinary immunology: principles and practice.* CRC Press.

Geha, R., & Notarangelo, L. (2012). *Case studies in immunology: a clinical companion.* Garland Science.

Jenni Punt; Sharon A Stranford; Patricia P Jones; Janis Kuby. (2013). *Kuby immunology*. New York: W.H. Freeman.

Joseph, A. Bellanti. (2016). *Immunology IV: Clinical Applications in Health and Disease*. Washington, DC: Georgetown University School of Medicine.

Murphy, K., & Weaver, C. (2016). *Janeway's immunobiology* (9th ed). Garland Science.

Rao, C. V. (2013). *Immunology* (2nd ed). New Delhi: Narosa Publishing House.

Outcomes	➢ Students acquire the information about immunity development ➢ Become an eminent in immunotechnology ➢ Able to understand the immunological reactions

Course Designed by: T. Kavitha

Semester - III			
Course code: 530303	**Industrial Microbiology**	**Credits:4**	**Hours: 4**

Objectives	➢ To give knowledge on strain improvement methods ➢ To learn about the upstream fermentation process ➢ To understand about the downstream fermentation process
Unit -I	**An introduction to fermentation process**:- Screening of industrial microbes – Detection and assay of fermentation products. Classification of fermentation types. Genetic control of fermentation. Strain selection and improvement, mutation - protoplast fusion, parasexual reproduction and recombinant DNA technique for strain development. Preservation methods of cultures.
Unit-II	**Types and design of bioreactors**:- Packed/fluidized, fed, transport phenomena – mass transfer, Newtonian and non – Newtonian behavior of fluid – mass transfer coefficient, oxygen, viscosity, heat transfer and scale up. Mode of operation. Instrumentation and computer application in fermentation
Unit III	**Fermentation kinetics**: Yield factors - growth rate parameters- kinetics of growth and product formation in batch, chemostat and fed-batch culture. Inoculum development, media formulation, optimization methods, media sterilization, statistical design for media formulation, optimization, and contour Plot. Immobilization of cells and enzymes - methods and applications.
Unit IV	**Fermentation of microbial products:**– Single Cell Protein (SCP).Anaerobic fermentation (beer and wine). Aerobic fermentation (vinegar and citric acid. Antibiotic fermentation (penicillin and streptomycin). Vitamins (B12, riboflavin), Hormone (gibberellic acid, IAA). Enzyme (amylase, protease). Biogas production.
Unit V	**Downstream processing:-** Cell disruption – physical and chemical methods. Precipitation. Filtration- batch and continuous filters. Centrifugation - types, liquid-liquid extraction, chromatography, membrane process, drying, crystallization. Quality control and evaluation of industrial products and packaging. Fermentation economics - market potential, process cost, recovery cost.

Reference and Textbooks:-

Casida, L.E.J.R. (2019). *Industrial Microbiology* (2nd ed). New Delhi: New Age International (P) Ltd., Publishers.

Crueger, W. (2017). *Biotechnology: A Test Book of Industrial Microbiology* (3rd ed), MEDTECH Publishers.

El-Mansi, E. M. T., Bryce, C. F. A., Arnold L. Demain and Allman, A.R. (2012). *Fermentation Microbiology and Biotechnology*, CRC Press.

Glick, B.R., and Patten, C.L. (2010). *Molecular Biotechnology Principles and Applications of Recombinant DNA* (4th ed). ASM Publishers.

Joshi, R.D. (2017). *Text Book of Industrial Microbiology*, Oxford.

Patel A.H. (2016). *Industrial Microbiology*. (2nd ed). New Delhi: Laxmi Publications (P) Ltd.

Peppler, H. and Pearman, D. (1979). *Microbial Technology, Vol.I*, New York: Academic Press.

Prescott, L.M., Harley, J.P. and Helin, D.A. (2015). *Microbiology* (5th ed). New Delhi: McGraw Hill.

Stanbury, P.F, Whitaker, A. and Hall, S.J. (2016). *Principles of Fermentation Technology* (3rd ed). New Delhi: Aditya Book (P) Ltd.

Waites, M.J., Morgan, N.L., Rockey, J.S. and Higton, G. (2001). *Industrial Microbiology: An Introduction.* London: Blackwell Science.

Outcomes	➢ Students will get knowledge on strain improvement ➢ Enable them to work in the fermentation industry ➢ Students will get an idea of the upstream and downstream fermentation process

Course Designed by: Dr V. Balasubramanian

Semester - III			
Course code:530304	**Lab V: Lab in Medical Microbiology**	**Credits:4**	**Hours/week: 6**

Objectives	➢ To provide technical knowledge on the collection and processing of clinicalsamples ➢ To prepare them to work in a clinical laboratory ➢ To learn the technique for isolation and identification of the pathogens

1. Staining methods for the morphological feature of pathogenic bacteria.
   A. Differential stains – Gram stain, Ziehl Neelsen's stain for AFB
   B. Cytological stains –
      i) Endospore stain – *Bacillus*, *Clostridium*
      ii Capsule stain – positive stain
   C. Stain for Amoeba / Intestinal protozoa / Malarial parasites – Ironhaematoxylin stain, Leishman's stain, Giemsa stain.
2. Diagnostic Bacteriology : Laboratory diagnosis (isolation & identification)
   i) Pyogenic infection- *Streptococci*
   ii) UTI infection – *E. coli, Proteus, Pseudomonas* and *Salmonella*
3. Kirby – Bauer disc diffusion technique.
4. Antimicrobial susceptibility testing by MIC and MBC
5. Electrophoretic separation of serum proteins.
6. Screening of antibiotic-producing microbes.
7. .Identification of dermatophytes from clinical samples
8. .Isolation, biochemical characterization and identification of the clinical pathogen from urine, pus, throat swab and sputum

Reference and Textbooks:

Bailey and Scott's Diagnostic Microbiology, (2006). London: Mosby.

Carpenter D.L.(1975). *Immunology and Serology* (3rd ed). London: W.B. Saunders Company.

Collins and Lyne's Microbiological methods, (2001). New York: Arnold publishers.

Demain, A.L, and Davis, J.E. (1999). *Manual of Industrial Microbiology and Biotechnology* (2nd ed). Washington: American Society for Microbiology.

Hudson, L. and Hay, F.C. (1989), *Practical Immunology* (3rd ed). Oxford: Blackwell Scientific Publications.

Lippincott Williams and Wilkins. Philadelphia, Baltimore (2006). *Koneman's Color Atlas and Text book of Diagnostic Microbiology.*

Noel R. Rose, Herman Friedman, John L. Fahey. (1986). *Manual of Clinical Laboratory Immunology*, American Society for Microbiology.

Patrick R. Murray, Ellen Jo Baron, James Jorgensen, Michael Pfaller, Marie Louise Landry. (2007). *Manual of Clinical Microbiology*: 2 Volume Set (9th Revised ed). American Society for Microbiology.

Rastogi S.C. (1996). *Immunodiagnostics Principles and Practice.* New Delhi: New Age International (P) Ltd.

Talwar, G.P. (1983). *A Handbook of Practical Immunology*. New Delhi: Vikas Publishing House Pvt. Ltd.

Outcomes	➢ Get practical knowledge in specimen collection and processing ➢ Become technically expert which will helpful to work in clinical laboratory ➢ Able to identify clinical pathogens

Course Designed by: Dr. T. Sathiamoorthi

Semester - III			
Course code: 530305	**Lab VI: Lab in Immunobiologyand Industrial Microbiology**	**Credits:4**	**Hours/week: 6**

Objectives	➢ To provide technical knowledge on the collection and processing of clinicalsamples ➢ To prepare them to work in a clinical laboratory ➢ To learn the technique for isolation and identification of the pathogens

1. Widal slide and tube agglutination test
2. Latex agglutination test- RA- test, CRP- test, ASO- test
3. Direct agglutination to determine ABO blood grouping.
4. Determination of differential leukocyte count.
5. Isolation and enumeration of RBC from human blood.
6. Demonstration of fermentation using Kuhn's fermentation vessel.
7. Assay of amylase from microbes
8. Assay of protease from microbes.
9. Assay of cellulase from microbes.
10. Enzyme Immobilization in sodium alginate gel.
11. Cell immobilization in calcium alginate gel

Reference and Textbooks:

Baily and Scott's Diagnostic Microbiology, (2006). London: Mosby.

Carpenter D.L.(1975). *Immunology and Serology* (3rd ed). London: W.B. Saunders Company.

Collins and Lyne's Microbiological methods, (2001). New York: Arnold publishers.

Demain, A.L, and Davis, J.E. (1999). *Manual of Industrial Microbiology and Biotechnology* (2nd ed). Washington: American Society for Microbiology.

Hudson, L. and Hay, F.C. (1989), *Practical Immunology* (3rd ed). Oxford: Blackwell scientific Publications.

Lippincott Williams and Wilkins. Philadelphia, Baltimore (2006). *Koneman's Color Atlas and Text book of Diagnostic Microbiology.*

Noel R. Rose, Herman Friedman, John L. Fahey. (1986). *Manual of Clinical Laboratory Immunology*, American Society for Microbiology.

Patrick R. Murray, Ellen Jo Baron, James Jorgensen, Michael Pfaller, Marie Louise Landry. (2007). *Manual of Clinical Microbiology*: 2 Volume Set (9th Revised ed). American Society for Microbiology.

Rastogi S.C. (1996). *Immunodiagnostics Principles and Practice.* New Delhi: New Age International (P) Ltd.

Talwar, G.P. (1983). *A Hand Book of Practical Immunology*. New Delhi: Vikas Publishing House Pvt. Ltd.

Outcomes	➢ Get practical knowledge in specimen collection and processing ➢ Become technically expert which will helpful to work in clinical laboratory ➢ Able to identify clinical pathogens

Course Designed by: Dr. T. Sathiamoorthi

Semester - III			
DSE -III: 533505	**Algal Biotechnology**	**Credits:3**	**Hours: 3**

Objectives	1. To learn about algal cultivation methods 2. To know the application of algae 3. To make aware of algae and pollution
Unit -I	**General overview, occurrence, and distribution of algae**:- Fundamentals of algal cultivation. Culture methods - batch cultures, continuous cultures, semi-continuous cultures, commercial-scale cultures, outdoor ponds, photobioreactors and culture of sessile microalgae. Quantitative determinations of algal density and growth, Growth rate and generation time determinations. Cultivation of economically important Fresh water and marine algae. Algae as a source of food and fodder.
Unit-II	**Algal Biotechnology**:- Application of cell fusion, tissue culture and hybridization techniques in algae. Algaegenomics. Genetic engineering of algae: construction of transformation and expression vectors, methods of gene introduction. Metabolic engineering in lipid metabolism. Phycoremediation. Role of algae in the biogeochemical cycle. Microalgal biotechnological applications in nutrition, health and environment.
Unit III	**Biofuels and Biofertilizer**:- Biogas, Ethanol, Diesel, and Hydrogen production by algae. Seaweed fertilizer and algae as Biofertilizer.
Unit IV	**Food and nutraceuticals of Algae**:- Cyanophyta, Rhodophyta, Heterokontophyta, Chlorophyta. Polysaccharides (Agar Agar, Carrageenan and Alginic acid), Algae in pharmaceutical industries, Animal feed, Therapeutic supplements and toxins.
Unit V	**Algae and pollution**:- Eutrophication, Algae as an indicator of pollution, atmospheric algae. Harmful algae blooms (HABS). Impacts of HABs on Aquaculture- Shellfish, Finfish, Impacts of Coastal HABs on Tourism.

Reference and Textbooks:-

Chapman, F.G. and Chapman, D.J. (1973). *The Algae*. McMillan & Co.

Faizal Bux , Yusuf Chisti (eds). (2016). *Algae Biotechnology- Products and Processes.* Springer International Publishing Switzerland.

Faizal Bux, Yusuf Chisti (eds.). (2018). *Algae Biotechnology_ Products and Processes- (Green Energy and Technology)* -Springer International Publishing.

Melanie N. Johansen. (2011). *Microalgae_ Biotechnology, Microbiology and Energy (Marine Biology)* --Nova Science Pub Inc.

Michael T. Madigan, John M. Martinko, Kelly S. Bender, Daniel H. Buckley, David A. Stahl, Thomas Brock. (2015). *Brock biology of microorganisms*-Benjamin Cummings.

Se-Kwon Kim. (2015). *Handbook of Marine Microalgae_* Biotechnology Advances. Academic Press.

Tridevi, P. C. (2001). Algal Biotechnology. Jaipur: Point Publisher.

VandenHoek, C., Mann, D.G., and Jahns, H.M. (2009). Algae- An introduction to Phycology.

Vashishta, B.R., Sinha, A.k., and Singh V.P. (2010). Algae (Revised) New Delhi: S.Chand & Company Ltd.

Williams, K.L. (2007).Endotoxins – Pyrogens, LAL Testing and Depyrogenation (3rd ed). INFORMA Publishers.

Outcomes	➢ Better understanding importance of algal biotechnology ➢ Get information about microalgae ➢ Enable them to work algal industries

Course Designed by: Dr.V. Balasubramanian

Semester - III			
DSE -III: 530506	**Applied Microbiology -I**	**Credits:3**	**Hours: 3**
Objectives	➢ To acquire knowledge on Applied Microbiology ➢ To enable them to know about the preservation of pharmaceutical products ➢ Learn to assess the microbial quality of marine foods.		
Unit -I	Microbiological standards I: General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025). International Standard on Medical Devices - Quality Management Systems - Requirements for Regulatory Purposes (ISO 13485).		
Unit-II	Microbiological standards II: Sterilization of health care products - Microbiological methods (BS EN ISO 11737-1:2018). Sterilization of medical devices - Microbiological Methods (BS EN ISO 11737-2:2009). Biological evaluation of medical devices - Tests for *in vitro* cytotoxicity- ISO 10993-5.		
Unit III	**Microbiology of Food technology**: Microbiological Risk Analysis Concepts: Risk assessment, risk management and risk communication; risk profiling of products. Bio-safety concepts in the handling of dairy/ food pathogens and setting up microbiological / pathogen lab in a dairy/food plant. Rapid Enumeration Techniques: Enumeration principles and procedure for rapid detection of predominant hygiene indicator organisms and pathogens like *E. coli*, *Salmonella*, *Shigella*, *Staphylococcus aureus*. *Bacillus cereus* and *Listeria monocytogenes*.		
Unit IV	**Pharmaceutical Microbiology:** The role of the Qualified Person in microbiological quality assurance, safety in Rapid microbiology enumeration and identification methods– selection and use of cleaning and disinfection agents in pharmaceutical manufacturing measurement of biocide effectiveness, International disinfectant testing protocols. The personal Qualification procedure for clean area entry –clean-in-Place, sterilization in-place, Cleanroom design, operation, and regulatory standards.		
Unit V	**Microbial nanotechnology:** Definition and terminologies- microbial nanotechnology, nanomedicine, nanowires, quantum Dots, nanocomposite, nanoparticles Synthesis of nanomaterial using microbes. Properties and characterization- imaging and Size and composition. Nanomaterials in diagnostics drug delivery and therapeutics. Nanomaterials – cytotoxicity and genotoxicity *–in vivo* test and assay. **Microbial biosensor:** Working mechanism and applications, advantages and limitations.		

Reference and Textbooks:-

Ashutosh, K. (2008). *Pharmaceutical Microbiology*. New Delhi: New Age InternationalPublishers.

Charalampopoulos, Dimitris, Rastall and Robert (2009), Prebiotics and Probiotics Science and Technology, Springer Publication

Doyle M. P. and Beuchat L. R. (2007). *Food Microbiology- Fundamentals and Frontiers*, ASM Press.

Kenji Sonomoto and Atsushi Yokota (2011), Lactic acid bacteria and Bifidobacteria, Caister Academic Press Publisher.

Kevin, W. (2007). *Endotoxins – Pyrogens, LAL Testing and Depyrogenation* (3rd ed). InformaPress.

Manivasakam, N. (2001). *Chemical and Microbial analysis of mineral and packaged drinkingwaters.* Coimbatore: Sakthi Book Service.

Peppler, H.J. and Pearlman, 2004. Microbial Technology, Vol – I, and Academic press, New Delhi.

Stanbury, P.F., Whitaker, A and Hall, S.A. 2000. Principles of Fermentation Technology, Second edition, Pergaman Press – Oxford.

Trivedy, R.K., Goel, P.K. and Trishal, C.L. (1987). *Practical methods in Ecology andEnvironmental science*. Environmental publishers.

Rajendran, P and P. Gunasekaran. (2007). *Microbial Bioremediation*. MJP. Publishers.

Outcomes	➢ Acquire Knowledge on food product analysis ➢ Impart knowledge of preservation technology. ➢ Knowledge on quality analysis of marine food products

Course Designed by: T. Kavitha

Semester - IV			
Course code: 530401	**Applied Microbiology -II**	**Credits:4**	**Hours: 5**
Objectives	➢ To acquire knowledge on food product analysis ➢ To enable them to know about the preservation of pharmaceutical products ➢ Learn to assess the microbial quality of marine foods.		
Unit -I	**Mineral water industry:** Stages of mineral water production. Analysis of water quality – pH, salinity, alkalinity, dissolved oxygen, carbonates, nitrate, silicate, phosphate, COD and BOD. Determination of microbial load in water: Faecal indicator organisms - coliform bacteria, fecal enterococci, Clostridium perfringens, yeast, mould and sulfide reducing anaerobes, viruses and bacteriophages, fungi and yeasts, protozoa and helminths. Methods of mineral water quality assessment – MPNtest, membrane filtration technique.		
Unit-II	**Preservation of pharmaceutical Products:** Chemical preservatives – raw materials – equipment – the role of preservatives. Finished product tests – microbial enumeration test, tests for specified microorganisms. Sterility testing – antimicrobial effectivenesstesting. Sterility assurance – biological indicators, sterilization validation process. Microbial risk assessment through HACCP plan.		
Unit III	**Endotoxin test methods:** gel clot assay, turbidimetric assay and chromogenic methods. Biological assays - vitamin assay, antibiotic susceptibility testing-Disc diffusion and well diffusion assay and mycoplasma testing. Endotoxin activity – risk assessment in parenteral manufacture – pyrogen test – de pyrogenation methods.		
Unit IV	**Rapid methods for detection of microorganisms in food**: conventional and automated. Application of light pulse technology – principles of light pulse generation, mode of action, equipment, application of light pulses, effect of light pulses on foods and microorganisms, advantage and limitations of light pulse treatment. Quality control in fruits and vegetable processing. Risk assessment in food industry – physical, chemical and biological hazards.		
Unit V	**Assessment of microbial quality of marine foods**: Conventional and recent development methods – flow cytometry, ATP estimation, radiometric, reflective colorimetry, LAL test, immunoassay, DNA based and microarray methods. Application of additives in food.Food safety and standard act for adulteration.Significance of barcode and its uses in the food industry.		

Reference and Textbooks:-

Ashutosh, K. (2008). *Pharmaceutical Microbiology*. New Delhi: New Age International Publishers.

John A. J. Barbara, Fiona A. M. Regan, Marcela Contreras. (2008). *Transfusion Microbiology*, United Kingdom: Cambridge University Press.

Joseph, A. Bellanti. (2016). *Immunology IV: Clinical Applications in Health and Disease*. Washington, DC: Georgetown University School of Medicine.

Kevin, W. (2007). *Endotoxins – Pyrogens, LAL Testing and Depyrogenation* (3rd ed). Informa Press.

Manivasakam, N. (2001). *Chemical and Microbial analysis of mineral and packaged drinking waters.* Coimbatore: Sakthi Book Service.

Michael J. Day, Ronald D. Schultz. (2014). *Veterinary Immunology*: *Principles and Practice* (2nd ed). CRC Press.

Raif Geha, Luigi Notarangelo. (2016). *Case Studies in Immunology*. *A Clinical Companion* (7th ed). ASM Press.

Rao, C. V. (2013). *Immunology* (2nd ed). New Delhi: Narosa Publishing House.

Richard Coico and Geoffrey Sunshine. (2015). *Immunology: A Short Course*, (7th ed). Wiley-Blackwell.

Trivedy, R.K., Goel, P.K. and Trishal, C.L. (1987). *Practical methods in Ecology and Environmental science*. Environmental publishers.

Course Designed by: T. Kavitha

Semester - IV			
Course code: 530999	**Project Report and Viva voce**	**Credits:10**	**Hours: 25**
Objectives	1. To inculcate knowledge on research 2. To know the principles of various instruments and its application in research 3. To familiarize in identifying a problem and to solve it.		
The students will do their research work under the research supervisor and submit their thesis.			
Outcomes	1. Knowledge on research methodology 2. Basic knowledge on different instruments 3. Ideas on research review		

Name of the Course Teacher: Research supervisor of the M.Sc candidate

Non-Major Electives Course (NME) (For II Semester) - To be chosen by other PG degree students:

Semester - II			
NME -I: 530507	**Molecular Biology**	**Credits:2**	**Hours: 3**
Objectives	➢ To extend the knowledge on the structure and functions of genetic materials ➢ To focus on genome organization, transcription and translation process in prokaryotes. ➢ To understand the principles of oncogenes		
Unit –I	**Discovery of DNA**. Molecular basis of DNA as genetic material. **Structure of DNA** – A, B and Z form. **Forms of DNA** – DNA heteroduplex, circular, superhelical DNA, twisted circle. **Properties of DNA** - denaturation, renaturation, melting curve, hyperchromicity. **Structure of RNA. Types of RNA** - tRNA, mRNA and rRNA.		
Unit-II	**Replication of DNA** - semi-conservative model, Meselson - Stahl experiment. **Enzymology of DNA replication** - DNA polymerase I, II and III; topoisomerase I and II; helicase; primase and gyrase. **Molecular mechanism of DNA replication**. Replication fork, origin and Okazaki fragments. **Types of replication** - circular and theta.		
Unit III	**Transcription process in Prokaryotes:** Initiation - promotors, upstream and downstream sequences, transcription factors; **Elongation** - RNA polymerase, subunits; **Termination** - Rho-dependent and Rho-independent; nus A protein and antitermination. **RNA processing** (post-transcriptional modifications), inhibitors of transcription. **Reverse transcription.**		
Unit IV	**Genetic code:** Elucidation of triplet code, code characteristics and codon dictionary. Reading frames, sense and nonsense code. **Degeneracy -** wobble hypothesis, the universality of genetic code. **Process of translation in prokaryotes:** Initiation and Termination. Role of rRNA in protein synthesis. **Post-translational modifications** - post-translational transport. Signal hypothesis.		
Unit V	**Tumor viruses and oncogenes:** Transformed cells, detection of integral viral DNA, the structure of integral viral DNA. Protein kinase and transformation by retroviruses. Cellular counterpart of src. Carcinogens. Activation of oncogenes. **Oncogenic proteins** - protein kinases, growth factors, ras protein. Transformation protein in DNA viruses.		

Reference and Textbooks:-

Benjamin Lewin. (2007). *Genes XI.* New York: Oxford University Press.

Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter. (2008). *Molecular Biology of the Cell* (5th ed). Garland Science.

David Freifelder. D. (2008). *Microbial Genetics* (18th ed). NewDelhi: Narosa Publishing House.

Freifelder, D. (2000). *Molecular Biology* (2nd ed). NewDelhi: Narosa Publishing house.

Jeyanthi, G.P. (2009). *Molecular Biology.* Chennai: MJP Publishers.

Stanley R. Maloy, John E.C. and Freifelder, D. (2008). *Microbial Genetics.* New Delhi: Narosa Publishing House.

Stryer, L. (2019). *Biochemistry* (9th ed). New York: W.H. Freeman and Company.

Veer Russel, P. (2009). *iGenetics: A Molecular Approach*. India: Pearson Education.

Watson, J.D., Hopkins, N.H., Roberts, J.W., Steitz, J.A. and Weiner, A. M. (2013). *Molecular Biology of the Gene* (17th ed). Tokyo: The Benjamin Cummings Publishing Company Inc.

Outcomes	➢ Receive elaborate knowledge on nucleic acids ➢ Better understanding of gene expressions ➢ Get thorough knowledge on tumour viruses and oncogenes

Course Designed by: Dr. A. Arun

Non-Major Electives Course (NME) (For II Semester) - To be chosen by other PG degree students:

Semester – II			
NME -I: 530508	**Agriculture and Environmental Microbiology**	**Credits:2**	**Hours: 3**
Objectives	➢ To create awareness on soil Microbiology ➢ To give knowledge on plant-pathogen interaction and its control ➢ To inculcate on environmental microbiology		
Unit -I	**Diversity and distribution of microorganisms in soil;** Soil Microflora- Bacteria, Fungi and Actinomycetes. Classification, physical, chemical properties and structure of soil. **Microbial interactions** - mutualism, synergism, commensalism, amensalism, parasitism, predation and competition. **Microbial interactions with plants**– phyllosphere, mycorrhizae, rhizosphere and symbiotic association in root nodules. **Biofertilizer** – VAM, *Rhizobium, Frankia, Azospirillum*, *Azotobacter*, Cyanobacteria, Phospho bacteria and *Azolla*.		
Unit-II	**Bacterial diseases of agricultural crops** - pathogens, symptoms, control measures with reference to paddy, cotton, maize, tomato, citrus, mango and potato. **Plant protection** – phenolics – phytoalexins and related compounds. Bioinsecticides – viral, bacterial and fungal- a brief note.		
Unit III	**Bio-geo chemical cycles in soil** – Carbon cycle, Nitrogen cycle – Nitrogen fixation, nitrification, denitrification, sulphur, iron and phosphorus cycles. **Aerobiology** – a brief introduction - droplet nuclei – aerosols - air-borne transmission of microbes and diseases and assessment of air quality.		
Unit IV	**Aquatic microbiology** - factors affecting microbial growth – temperature – pressure – light – salinity - turbidity – pH -inorganic and organic constituents. **Aquatic habitats** - freshwater - lakes, ponds and streams; **marine habitats** - estuaries, deep sea, hydrothermal vents, saltpans, coral reefs and mangroves and their microbial communities; **zonation** – food chain and food web.		
Unit V	**Types of wastes** - solid and liquid wastes. **Treatment of solid wastes** - Thermal Treatment: Incineration, Gasification, Pyrolysis and Open Burning- **Dumps and Landfills**: Sanitary landfills, Controlled dumps, Bioreactor Landfills-**Biological Waste Treatment**: Composting, Vermicomposting and termi composting. **Treatment of liquid wastes** –primary, secondary, tertiary treatment; anaerobic (methanogenesis), aerobic, trickling, activated sludge, oxidation pond. Production of biogas from waste.		

Reference and Textbooks:-	
Alexander M. (1997). *Introduction to soil microbiology*, New York: John Wiley & Sons, Inc. EcEldowney S., Hardman, D.J. and Waite, S. (1993). *Pollution Ecology and Biotreatment*. Longman Scientific Technical. Grant, W.D. and Long, P.L. (1981). *Environmental Microbiology*. Blalckie Glasgow and London. Madigan, M.T., Martinka, M., Parker, J. and Brock, T.D. (2000). Twelfth Edition, *Biology Microorganisms*, New Jerry: Prentice Hall. Mark Wheelis, (2010). *Principles of Modern Microbiology*, New Delhi: Jones & Bartlett India Pvt. Ltd. Mehrotra, R.S. (1983). *Plant Pathology*, New Delhi: Tata McGraw Hill Publishing Company Ltd. Pandy, B.P. (1997). *Plant Pathology (Pathogen & Plant Disease)*, New Delhi: S.Chand& Company Ltd. Ray Chadhuri, S.P. (1977). *A Manual of Virus Diseases of Tropical Plants*, New Delhi: MacMillan Company of India Ltd. Rengaswami, G. and Rajagopalan, S. (1973). *Bacterial Plant Pathology*. Coimbatore: Tamil Nadu Agriculture University. SubbaRao, N.S. (1995). *Soil Microorganisms and Plant Growth* (3rd ed). New Delhi: Oxford & IBH Publishing Co. Pvt. Ltd.	
Outcomes	➢ Acquire knowledge on soil microbiology ➢ Understand the biogeochemical cycles prevail in the environment. ➢ Able to know about principles and techniques in waste treatment.

Course Designed by: Dr. T. Sathiamoorthi

Non-Major Electives Course (NME) (For III Semester) - To be chosen by other PG degree students:

Semester – III			
NME-II: 530509	**Medical Microbiology**	**Credits:2**	**Hours: 3**

Objectives	➢ To inculcate on the role of normal flora and pathogenic microbes ➢ To understand the pathogenesis of various diseases ➢ To understand the various clinical, microbiological techniques
Unit -I	**Laboratory management: Normal flora of human systems** – skin, respiratory tract, gastrointestinal tract and genitourinary tract. Nosocomial infections. Collection, transport of clinical samples and laboratory waste disposal system. Microbiological examination of urine, blood, feces, cerebrospinal fluid, throat swabs, sputum, pus and wound exudates.
Unit-II	**Bacterial Diseases:** Morphology, classification, cultural characteristics, pathogenicity, laboratory diagnosis and prevention of infections caused by the following organisms: **Gram positive cocci**– *Staphylococci, Streptococci.* **Gram negative cocci**– *Gonococci.* **Gram positive non-spore-forming bacilli**: aerobic – *Corynebacteria* and anaerobic
Unit III	**Fungal Disease:** General characteristics, pathogenesis and laboratory diagnosis and control measures of: **Yeast**–*Cryptococcus neoformans.* **Yeast-like fungus** – *Candida spp*. **Filamentousfungi** – *Aspergillus* and *Penicillium.* Dimorphic fungi, yeast morphology, general characteristics and reproduction. – *Blastomyces dermatitis.*
Unit IV	**Viral Disease:** Infectious diseases- Definition of emerging & re-emerging diseases. Factors contributing to emergence. Examples (Chickungunya, Zika virus, H1N1 and Ebola). National programs in the prevention of infectious diseases.
Unit V	**Parasitic Diseases:** General characteristics of Intracellular parasites– *Cryptosporidium* and *Plasmodium.* Intralumen parasites – *Entameoba histolytica* and *Ascaris lumbricoides.* Parasitic zoonosis– *Toxoplasma* and *Taenia*

Reference and Textbooks:

Anathanarayan R and Jeyaram Panikers C.K. (2013). *Text Book of Microbiology* (9th ed). New Delhi: Jain book depot.

Arora D.R., Brij Bala Arora.(2015). *Textbook of Microbiology*. Chennai: CBS.

Awetz Melnick and Adelberg's. (2010). *Medical Microbiology, 21st Century.* Appleton & Lange.

Bhattacharjee R.N.(2015). *Introduction to Microbiology* (1st ed). New Delhi: Kalyani Publishers.

Connie R Mahon. (2010). *Textbook of Diagnostic Microbiology* (3rd ed). Pearson.

David Greenwood, Richard Slack, John Peutherer. (2012). *Medical Microbiology.* Churchill Livingstone.

Jesse Russell, Ronald Cohn. (2012). *Medical Microbiology.* Book on Demand Ltd.Myra

Patrick R. Murray. (2015). *Medical Microbiology.* Elsevier

Patrick Murray & Ken Rosenthal & Michael Pfalle. (2015). *Medical Microbiology* (8th ed). New York: Academic Press.

Wilkinson. (2011). *Medical Microbiology.* Scion Publishing Ltd

Outcomes	➢ understand the normal flora of human body ➢ know about the clinical technology ➢ understand the human pathogen

Course Designed by: Dr. T. Kavitha

Non-Major Electives Course (NME) (For III Semester) - To be chosen by other PG degree students:

Semester – III			
NME -II: 530510	Food and Industrial Microbiology	**Credits:2**	**Hours: 3**
Objectives	➢ To learn the diversity of microbes in the fermentation of food ➢ To get knowledge of food-borne diseases ➢ To learn about fermentation types ➢ To understand the extraction and purification of fermented products		
Unit -I	**History and development of Food microbiology:** Role and Significance of Microorganisms in Foods, Outline of food spoilage and preservation, Food safety and microbiological Quality Assurance. Production of fermented dairy products**:** Cheese, yogurt, buttermilk, sourcream, Fermented vegetables; Sauerkraut.		
Unit-II	**Food spoilage, preservation and foodborne diseases:** organism involved in spoilage of fruits and vegetables, cereal and cereal products, meat and meat products. Foodborne diseases- Bacterial foodborne diseases- (Staphylococcal intoxication, Botulism, Salmonellosis, Shigellosis, EPEC Diarrhoea,; Food-borne fungi- Mycotoxins-Aflatoxicosis.		
Unit III	**An introduction to fermentation process**:- Screening of industrial microbes – Detection and assay of fermentation products. Classification of fermentation types. Inoculum development, media formulation, optimization methods, media sterilization, statisticaldesign for media formulation, and optimization.		
Unit IV	**Fermentation of microbial products:**– Single Cell Protein (SCP). Anaerobic fermentation (beer and wine). Aerobic fermentation (vinegar and citric acid. Antibiotic fermentation (penicillin and streptomycin). Vitamins (B12, riboflavin), Hormone (gibberellic acid, IAA). Enzyme (amylase, protease). Biogas production.		
Unit V	**Downstream processing:-** Cell disruption – physical and chemical methods. Precipitation. Filtration- batch and continuous filters. Centrifugation - types, liquid-liquid extraction, chromatography, membrane process, drying, crystallization.		

Reference and Textbooks:-

Adams, M.R. and Moss, M.O. (2008). Food Microbiology. UK: RSC Publishing, Cambridge. Aneja, K.N. (2018). Modern Food Microbiology, Medtec Publisher.

Casida, L.E.J.R. (2019). *Industrial Microbiology* (2nd ed). New Delhi: New Age International (P) Ltd., Publishers.

Crueger, W. (2017). *Biotechnology: A Test Book of Industrial Microbiology* (3rd ed), MEDTECH Publishers.

Demain, A.L. and Davis, J.E.2004, Industrial Microbiology and Biotechnology, (2/e), ASM Press Washington, DC.

Dick M, (2017). Food Microbiology An Introduction (2nd ed). Bengaluru: Medtech.

El-Mansi, E. M. T., Bryce, C. F. A., Arnold L. Demain and Allman, A.R. (2012). *Fermentation Microbiology and Biotechnology*, CRC Press.

Frazier, W.C., and Westhoff, D.C. (1988). Food Microbiology (Reprint 1995). New Delhi: Tata McGraw Hill Publishing Ltd.

Stanbury, P.F, Whitaker, A. and Hall, S.J. (2016). *Principles of Fermentation Technology* (3rd ed). New Delhi: Aditya Book (P) Ltd.

Prescott, L.M., Harley, J.P. and Helin, D.A. (2015). *Microbiology* (5th ed). New Delhi: McGraw Hill.

Outcomes	➢ Get knowledge on importance of microbes fermentation of food ➢ Students will get knowledge on strain improvement ➢ Enable them to work in fermentation industry ➢ Students will get idea on upstream and downstream fermentation process

Course Designed by: T. Kavitha

CURRICULUM VITAE

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Educational qualification:

- M.Sc., Microbiology
- M.Phil, Biotechnology
- Ph.D., Microbiology

Professional experience:

- 20 years of Teaching and Research Experience

Honors and Awards:

1. Young scientist award – DST-SERB-YSS-2013
2. Dr. APJ Abdul Kalam Award for scientific excellence

Recent publications:

1. K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh, Kunyu Zhang, Sudhakar Muniyasamy, A. Pugazhendhi, J. Jeyakanthan, Kumar Ponnuchamy, M. Govarthanan, A. Arun. 2021. Production and Characterization of biodegradable Polyhydroxybutyrate by *Micrococcus luteus* isolated from marine environment. International Journal of Biological Macromolecules. (IF – 8.025). DOI: https://doi.org/10.1016/j.ijbiomac.2021.07.029
2. Premnath. N, K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh, G. Siva Prakash, V. Ananthi, Kumar Ponnuchamy, Govarthanan Muthusamy, A. Arun. (2021). A Crucial Review on Polycyclic Aromatic Hydrocarbons - Environmental Occurrence and Strategies for Microbial Degradation. Chemosphere. DOI: 10.1016/j.chemosphere.2021.130608 (IF – 8.943).
3. V. Ananthi, P. Balaji, Raveendran Sindhu, Sang-Hyoun Kim, Arivalagan Pugazhendhi , A. Arun. (2021). A critical review on different harvesting techniques for algal based biodiesel production. Science of the Total Environment. 78; 146467 DOI: https://doi.org/10.1016/j.scitotenv.2021.146467 (IF – 10.753).
4. Boobalan Thulasinathan, Tamilmani Jayabalan, Murugan Sethupathi, Woong Kim, Sudhakar Muniyasamye, Nallathambi Sengottuvelan, Samsudeen Nainamohamed, Kumar Ponnuchamy, Arun Alagarsamy. (2021). Bioelectricity Generation by Natural Microflora of Septic Tank Wastewater (STWW) and Biodegradation of Persistent Petrogenic Pollutants by

Basidiomycetes Fungi: An Integrated Microbial Fuel Cell System. Journal of Hazardous Materials. Volume 412, 25228, ISSN 0304-3894. https://doi.org/10.1016/j.jhazmat.2021.125228 (IF – 14.224).

5. Premnath. N, K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh , G. Siva Prakash, Arivalagan Pugazhendhi, J. Jeyakanthan, Muthusamy Govarthanan, Ponnuchamy Kumar, A. Arun. 2020. Effect of C/N Substrates for enhanced Extracellular Polymeric Substances (EPS) Production and Poly Cyclic Aromatic Hydrocarbons (PAHs) degradation. Environmental Pollution. 116035, ISSN 0269-7491 (IF – 9.988). DOI: https://doi.org/10.1016/j.envpol.2020.116035
6. V. Ananthi , Arivalagan Pugazhendhi , A. Arun. Efficacy of chemical factors on production and extraction of biodiesel by Microalgae. 2020. International Journal of Energy Research. (IF – 5.164). DOI: https://doi.org/10.1002/er.6097
7. . V Ananthi, Kathirvel Brindhadevi, Arivalagan Pugazhendhi, Arun Alagarsamy. 2020. Impact of abiotic factors on biodiesel production by microalgae. Fuel. https://doi.org/10.1016/j.fuel.2020.118962 (IF – 8.035).
8. Ramu Satheesh Murugan, Gujuluva Hari Dinesh, Thulasinathan Boobalan, Thondi Rajan Angelin Swetha, Ponnuchamy Kumar, Arivalagan Pugazhendhi, Alagarsamy Arun. 2020. Dark fermentative biohydrogen production from rice mill wastewater. International Journal of Energy Research. http://dx.doi.org/10.1002/er.5829 (IF – 5.164).
9. V Ananthi, Rathinam Raja, Isabel S Carvalho, Kathirvel Brindhadevi, Arivalagan Pugazhendhi; Arun Alagarsamy, 2020. A realistic scenario on microalgae-based biodiesel production: Third generation biofuel. Fuel. https://doi.org/10.1016/j.fuel.2020.118965 (IF – 8.035).
10. Ramu Satheesh Murugan, Gujuluva Hari Dinesh, Ramalingam Karthik Raja, Ebenezer Samuel James Obeth, Abhispa Bora, Naina Mohammed Samsudeen , Arivalagan Pugazhendhi, Alagarsamy Arun. 2020. Dark fermentative biohydrogen production by *Acinetobacter junii*-AH4 utilizing various industry wastewaters. International Journal of Hydrogen Energy. https://doi.org/10.1016/j.ijhydene.2020.07.073 (IF: 7.139).

Cumulative Impact factor: **362.063**

Total Citation: **1411**

h- index:**23**

i10- index: **36**

CURRICULUM VITAE

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Educational qualification:

- M.Sc., Microbiology
- M.Phil., Biotechnology
- Ph.D., Microbiology

Professional experience:

- 15 years of Teaching Experience
- 7 years of research Experience

Honours and Awards:

- Early Career Research Award from DST/SERB

Recent publications:

1. Karuppiah, V., Natarajan, S., Gangatharan, M., Munirah, F.A., Noorah, A. and **Thangavel, K**. (2022), Development of siderophore based rhizobacterial consortium for the mitigation of biotic and abiotic environmental stresses in tomatoes: An *in vitro and in planta approach*. *Journal of Applied Microbiology*, (Published Online on 17.05.2022), https://doi.org/10.1111/jam.15625 [Press: Wiley; Scopus indexed & SCIE indexed; **IF: 3.77**].
2. Vijay Karuppiah and **KavithaThangavel** (2022)**.** Inoculation of Medicinal Plant Resident *Staphylococcus hominis subsp. novobiosepticus* Strain KE6 for Improved Plant Growth and Antagonism to *Pithomycesatro-olivaceous* in Groundnuts. *Asian Journal of Research in Biosciences 4*(1): 12-25
3. Nagarajan Arumugam, Thulasinathan Boobalan, Arivalagan Pugazhendhi, Alagarsamy Arun, Muthuramalingam Jothi Basu, Thangarasu Suganya Devi, **Thangavel Kavitha**. "Particle size influence on the composition of sugars in corncob hemicellulose hydrolysate for xylose fermentation by *Meyerozyma caribbica*". ***Bioresource Technology***, 2021, 125677. **(IF: 9.64)**
4. Devi, T. S., Vijay, K., Vidhyavathi, R. M., Kumar, P., Govarthanan, M., and **Kavitha, T**. (2021). Antifungal activity and molecular docking of phenol, 2, 4-bis (1, 1-dimethylethyl)

produced by plant growth-promoting actinobacterium *Kutzneria sp*. strain TSII from mangrove sediments. Arch Microbiol, 1-14. https://doi.org/10.1007/s00203-021-02397-1 (Indexing: Scopus; WoS; Impact Factor: **2.5**; Springer-Verlag, Germany; NAAS score: 7.64).

5. Vijay, K., Kiran, G. S., Divya, S., Thangavel, K., Thangavelu, S., Dhandapani, R., & Selvin, J. (2021). Fatty Acid Methyl Esters From the Coral-Associated Bacterium Pseudomonas aeruginosa Inhibit Virulence and Biofilm Phenotypes in Multidrug Resistant *Staphylococcus aureus*: An *in vitro* Approach. *Front Microbiol*, 12, 394. https://doi.org/10.3389/fmicb.2021.631853 (Indexing: Scopus; WoS; Open Access; Impact Factor:**4.23**; Frontiers, USA; NAAS Score: 10.26)
6. Karuppiah Vijay, Thangarasu Suganya Devi, Karthikeyan KirupaSree, Abdallah M. Elgorban, Ponnuchamy Kumar, Muthusamy Govarthanan and **Thangavel Kavitha** (2020). *In vitro* screening and *in silico* prediction of antifungal metabolites fromrhizobacterium*Achromobacterkerstersii* JKP9. *Arch Microbiol*., 202: 2855-2864. (Springer-Verlag; IF: **2.5**; NAAS score: 7.64) https://doi.org/10.1007/s00203-020-01982-0.

Cumulative Impact factor: **24.5**

Total Citation: **113**

h- index:**7**

i10- index: **5**

CURRICULUM VITAE

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Educational qualification:

- M.Sc.,
- M.Phil.,
- Ph.D.,

Professional experience:

- Seven years

Recent publications:

1. Ajucarmel precilla, Arulprakasam, Jhansi Pandi, Ranjithkumar Dhandapani, Saikishore Ramanathan, Jayaprakash Chinnappan, Ragul Paramasivam, ***Sathiamoorthi Thangavelu*** et al. "In Silico Identification of Hub Genes as Observing Biomarkers for Gastric Cancer Metastasis." *Evidence-Based Complementary and Alternative Medicine* 2022 (2022).
2. ***Thangavelu Sathiamoorthi***, Ranjithkumar Dhandapani, Ajucarmelprecilla Arulprakasam, Ragul Paramasivam, Arunachalam Chinnathambi, Sulaiman Ali Alharbi, Kaliannan Durairaj, and Anupama Shrestha. "Isolation, Identification, Characterization, and Plasmid Profile of Urinary Tract Infectious Escherichia coli from Clinical Samples." *Evidence-Based Complementary and Alternative Medicine* 2022 (2022).
3. Pandi, Jhansi, Ajucarmelprecilla Arulprakasam, Ranjithkumar Dhandapani, Saikishore Ramanathan, ***Sathiamoorthi Thangavelu***, Jayaprakash Chinnappan, V. Vidhya Rajalakshmi, Saad Alghamdi, Nashwa Talaat Shesha, and S. Prasath. "Biomarkers for Breast Adenocarcinoma Using In Silico Approaches." *Evidence-Based Complementary and Alternative Medicine* 2022 (2022).
4. Rahuman, Haajira Beevi Habeeb, Ranjithkumar Dhandapani, Velmurugan Palanivel,***Sathiamoorthi Thangavelu***, Ragul Paramasivam, and Saravanan Muthupandian. "Bioengineered phytomolecules-capped silver nanoparticles using Carissa carandas leaf extract to embed on to urinary catheter to combat UTI pathogens." *PloS one* 16, no. 9 (2021): e0256748.

Cumulative Impact factor:

Total Citation: 110**h-index**:5

i10-index

CURRICULUM VITAE

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Karaikudi-630003,Tamil Nadu, INDIA
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microbaalu82@gmail.com

Educational qualification:

- M.Sc., Ph.D.

Professional experience:

- Six Years

Research Experience:

- Thirteen years

Honours and Awards:

- "Science Academies' Summer Research Fellowship Programme for Students and Teachers 2017". Indian Academy of Science, Bengaluru, India
- Recipient of JRF & SRF, University Grants Commission Fellowship, New Delhi (13.10.2009 to 29.01.2014).
- Received Best Paper Award for ORAL PRESENTATION in *"National Seminar on Coastal Typology"* held during March 15–17, 2010 at Department of Marine Science, Bharathidasan University – 620024. Title of the Paper "Biofouling on Marine Engineering Alloys"

Recent publications:

1. ***Balasubramanian, V**.** R Rajaram, S. Palanichamy, G Subramanian, K Mathivanan, A Pugazhendhi (2018) Lanosterol expressed bio-fouling inhibition on Gulf of Mannar coast, India. Journal of Progress in Organic Coatings (115)100–106, Elsevier, (IF-4.5)
2. Mathivanan, K., R. Rajaram, ***V. Balasubramanian**** and G. Annadurai, (2016), "Removal of Cd(II) and Cu(II) from Aqueous Solutions by *Pseudomonas stutzeri* KMNTT-01 Biomass. Environ. Process, 3(4) 857-874,
3. ***Balasubramanian, V.*,*** R. Rajaram and P. Rajesh (2016) Antifouling Potentials of

Mangrove *Rhizophora apiculata* against *Vibrio alginolyticus* strain VBM-01. Indian Journal of Geo-Marine Science, 44(10) 1654-1661 (IF-0.5), ISSN:0379-5136.

4. Mathivanan, K., R. Rajaram and ***V., Balasubramanian**** (2015) Biosorption of Cd(II) and Cu(II) ions using *Lysinibacillus fusiformis* KMNTT-10:equilibrium and kinetic studies, Desalination and Water Treatment,57 (47),22429-22440 doi:10.1080/19443994.2015.1129508(IF-1.6).
5. Sengol Jenifer, S. ***Balasubramanian V****. and R. Rajaram (2014) Chemical Characterization and Bioactivity Evaluation of Bacteriocin from Biofilm Forming Bacteria. African Journal of Microbiology Research,8(41),3617-3624, (ISSN:19960808).https://doi.org/10.5897/AJMR2014.7022

Cumulative Impact factor: **11.5**

Total Citation: **113**

h-index: **5**

i10-index: **2**

CURRICULUM VITAE

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Educational qualification:

- M.Sc.,
- PG Diploma.
- Ph.D.

Professional experience:

- 2 years of Teaching and 8 years of Research Experience

Recent publications:

1. Sivaprakash G, Mohanrasu K, Dinesh GH, Ananthi V, Boobalan T, Jothibasu M, Arun A et al. Environmental friendly synthesis of TiO_2-ZnO nanocomposite catalyst and silver nanomaterials for the enhanced production of biodiesel from *Ulva lactuca* seaweed and potential antimicrobial properties against the microbial pathogens. J Photochem Photobiol B Biol 2019;193:118–30. doi:10.1016/j.jphotobiol.2019.02.011. Impact factor: 6.814
2. Sivaprakash G, Mohanrasu K, Ananthi V, Jothibasu M, Nguyen DD, Ravindran B, Arun A et al. Biodiesel production from *Ulva linza, Ulva tubulosa, Ulva fasciata, Ulva rigida, Ulva reticulate* by using Mn_2ZnO_4 heterogenous nanocatalysts. Fuel 2019;255:115744. doi:10.1016/j.fuel.2019.115744. Impact factor: 8.035
3. Sivaprakash G, Mohanrasu K, Obeth J, Bora A, Yuvakkumar R, Hossam A, Arun A et al. Zinc based iron mixed oxide catalyst for biodiesel production from *Entermorpha intestinalis* , *Caulerpa racemosa* and *Hypnea musicoformisis* and antibiofilm analysis using leftover catalyst after transesterification. J King Saud Univ - Sci 2020. doi:10.1016/j.jksus.2019.12.018. Impact factor: 4.011
4. Sivaprakash G, Mohanrasu K, Ravindran B, Jin Chung W, Al Farraj DA, Soliman Elshikh M, Arun A et al. Integrated approach: Al_2O_3-CaO nanocatalytic biodiesel production and antibacterial potential silver nanoparticle synthesis from Pedalium murex extract. J King Saud Univ - Sci 2019. doi:10.1016/j.jksus.2019.12.004. Impact factor: 4.011
5. Ananthi V, Siva Prakash G, Mohan Rasu K, Gangadevi K, Boobalan T, Raja R, Arun A et al. Comparison of integrated sustainable biodiesel and antibacterial nanosilver production by

microalgal and yeast isolates. J Photochem Photobiol B Biol 2018;186:232–42. doi:10.1016/j.jphotobiol.2018.07.021. Impact factor: 6.814

6. Ananthi V, Siva Prakash G, Chang SW, Ravindran B, Nguyen DD, Vo D-VN, Arun A et al. Enhanced microbial biodiesel production from lignocellulosic hydrolysates using yeast isolates. Fuel 2019;256:115932. doi:10.1016/j.fuel.2019.115932. Impact factor: 8.035
7. Mohanrasu K, Rao, R Guru Raj, Dinesh G. H, Zhang Kunyu, Siva Prakash G, Dong Po Song, Arun A, et al. Optimization of media components and culture conditions for polyhydroxyalkanoates production by *Bacillus megaterium.* Fuel 271, 117522. https://doi.org/10.1016/j.fuel.2020.117522. Impact factor: 8.035
8. Dinesh GH, Nguyen DD, Ravindran B, Chang SW, Vo D-VN, Bach Q-V, Siva Prakash G, Arun A et al. Simultaneous biohydrogen (H_2) and bioplastic (poly-β-hydroxybutyrate-PHB) productions under dark, photo, and subsequent dark and photo fermentation utilizing various wastes. Int J Hydrogen Energy 2019. doi:10.1016/j.ijhydene.2019.09.036. Impact factor: 7.139
9. Mohanrasu K, Premnath N, Siva Prakash G, Sudhakar M, Boobalan T, Arun A. Exploring multipotential uses of marine bacteria; an integrated approach for PHB production, PAHs and polyethylene biodegradation. J Photochem Photobiol B Biol 2018;185:55–65. doi:10.1016/j.jphotobiol.2018.05.014. Impact factor: 6.814
10. Premnath, N., Mohanrasu, K., Raj Rao, R.G., Dinesh, G.H., Sivaprakash, G., Pugazhendhi, A., Jeyakanthan, J., Govarthanan, M., Kumar, P., Arun, A., 2020. Effect of C/N Substrates for enhanced Extracellular Polymeric Substances (EPS) Production and Poly Cyclic Aromatic Hydrocarbons (PAHs) degradation. Environ. Pollut. 703, 116035. https://doi.org/10.1016/j.envpol.2020.116035. Impact factor: 9.988
11. Premnath, N., Mohanrasu, K., Guru Raj Rao, R., Dinesh, G. H., Prakash, G. S., Ananthi, V., Ponnuchamy, K., Muthusamy, G., & Arun, A. (2021). A crucial review on polycyclic aromatic Hydrocarbons - Environmental occurrence and strategies for microbial degradation. *Chemosphere*, *280*(March), 130608. https://doi.org/10.1016/j.chemosphere.2021.130608. Impact factor: 8.943

Cumulative Impact factor: **78**

Total Citation: **260**

h- index:**8**

i10- index**: 8**

CURRICULUM VITAE

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Educational qualification:

- M.Sc., Microbiology
- Ph.D., Biotechnology

Professional experience:

- 07 years of Teaching and Research Experience

Honors and Awards:

- Awarded **"Young Scientist Award"** from Department of Science and Technology (DST), **Government of India,** New Delhi, **India**
- Awarded **"Travel Grant**" from Department of Biotechnology **(DBT),** Govt. of India, New Delhi to attend an International conference in **Canada**
- Awarded **"Travel Grant**" from **CICS**, Govt. of India, Chennai to attend an International conference in **Canada**
- Awarded **"Research Fellowship for Foreign Citizens"** funded by Scientific and Technological Research Council of Turkey (TUBITAK), Government of Turkey
- Awarded **"Turkish Government Scholarships 08-09"** by **Govt. of India** and **Govt. of Turkey Student exchange program**
- Awarded "**Project Fellow**" sponsored by University Grants Commission, Govt. of India

Recent publications:

1. **R. Karthik Raja**, P. Nguyen-Tri, G. Balasubramani, A. Arun, S. Hazir, S. Ladhari, A. Saidi, A. Pugazhendhi, A. Anthoni Samy. (2021). SARS-CoV-2 and its new Variants: A Comprehensive Review on Nanotechnological Application Insights into Potential Approaches. Applied Nanoscience. **(Impact Factor: 3.674)**

2. **R. Karthik Raja**, A. Arun, M. Touray, S.H. Gulsen, H. Cimen, B. Gulcu, C. Hazir, D. Aiswarya, D. Ulug, I. Cakmak, H. K. Kaya, S. Hazir. (2021). Antagonists and defense mechanisms of entomopathogenic nematodes and their mutualistic bacteria. Biological Control. **(Impact Factor: 3.687)**

3. A. Priya, A. Selvaraj, D. Divya, **R. Karthik Raja** and S. Karutha Pandian. (2021). In Vitro and In Vivo Anti-infective Potential of Thymol against Early Childhood Caries Causing Dual Species *Candida albicans* and *Streptococcus mutans*. Front. Pharmacol. 12:760-768. **(Impact Factor: 5. 81)**

4. Arul D., Balasubramanian V., Balasubramani G., **Karthik Raja R.,** Aiswarya D. and Perumal P. (2022). Isolation of a bacterial strain from the gut of the fsh, Systomus sarana, identifcation of the isolated strain, optimized production of its protease, the enzyme purifcation, and partial structural characterization. Journal of Genetic Engineering and Biotechnology.

5. A. Pugazhendhi, S. Vasantharaj, S. Sathiyavimal, **R. Karthik Raja**, I. Karuppusamy, M. Narayanan, S. Kandasamy, K. Brindhadevi. (2021). Organic and inorganic nanomaterial coatings for the prevention of microbial growth and infections on biotic and abiotic surfaces. Surface and Coatings Technology, 425, 127739. **(Impact Factor: 4.158)**

6. R. S. Murugan, G. H. Dinesh, **R. K. Raja**, E. S. J. Obeth, A. Bora, N. M. Samsudeen, A. Pugazhendhi, A.Arun. (2020). Dark fermentative biohydrogen production by Acinetobacter junii-AH4 utilizing various industry wastewaters. International Journal of Hydrogen Energy. 46, 11297-11304 **(Impact Factor: 5.816).**

7. D. Aiswarya, **R. K. Raja**, C. Kamaraj, G. Balasubramani, P. Deepak, D. Arul, V. Amutha, C. Sankaranarayanan, S. Hazir, P. Perumal. (2019). Biosynthesis of Gold and Silver Nanoparticles from the Symbiotic Bacterium, *Photorhabdus luminescens* of Entomopathogenic Nematode: Larvicidal Properties against Three Mosquitoes and *Galleria mellonella* Larvae. *Journal of Cluster Science*. **(Impact Factor: 3.061).**

8. D. Aiswarya, **R. Karthik Raja**, C. Kamaraj, G. Balasubramani, P. Deepak, D. Arul, P. Perumal (2018). Isolation and growth inhibition potential of entomopathogenic nematodes against three public health important mosquito vectors. *Experimental Parasitology.* 197, 76-84. **(Impact Factor: 2.011).**

9. Balasubramani, G., P. Deepak, D. Aiswarya, **R. Karthik Raja**, C. Kamaraj & P. Perumal (2018). Multipurpose efficacy of the lyophilized cell-free supernatant of Salmonella bongori isolated from the freshwater fish, Devario aequipinnatus: toxicity against microbial pathogens and mosquito vectors. *Environmental Science and Pollution Research*. **(Impact Factor: 4.223).**

Cumulative Impact factor: **60.05**

Total Citation: **289**

h- index:**11**

i10- index: **13**